Theoretical Ecotoxicology
Session 3: Toxicity prediction

University Duisburg-Essen SS 2026

Ralf B. Schäfer

Ecotoxicology, Faculty of Biology, UDE & RC One Health Ruhr

Intended learning outcomes of today's lecture

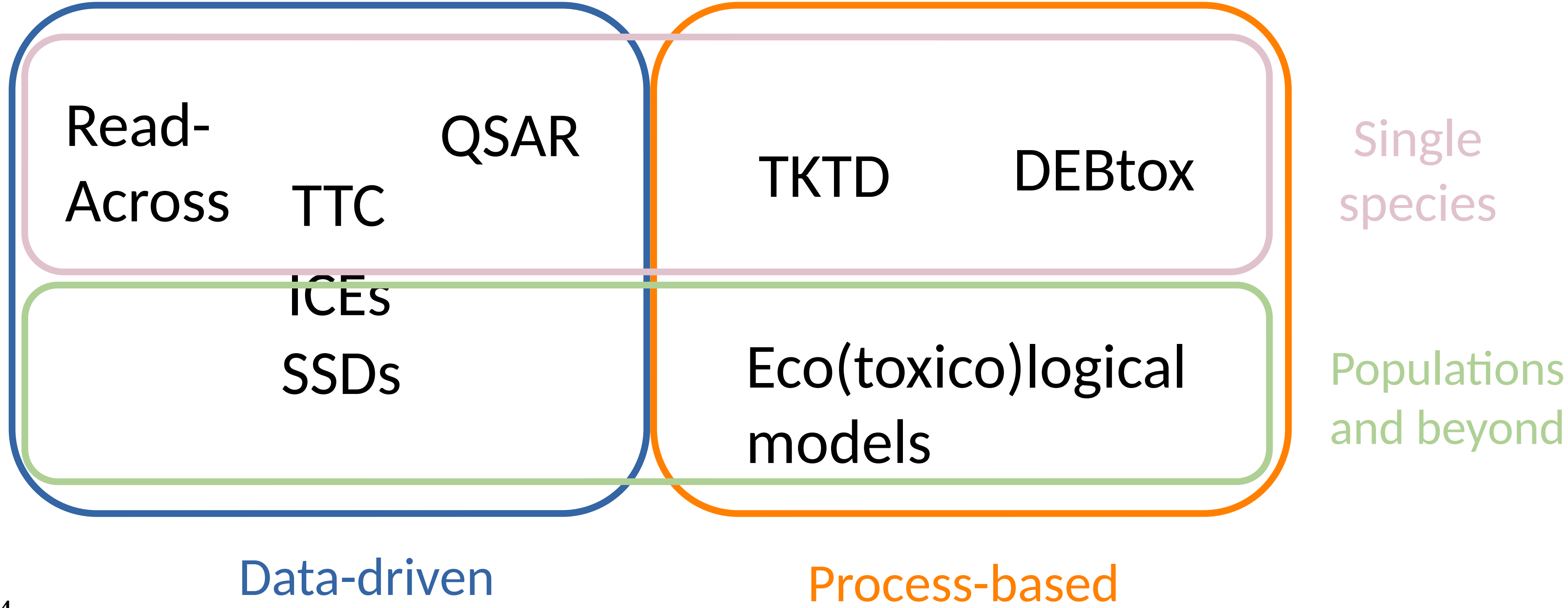
- Understanding the major concepts for toxicity prediction
- Knowledge on pre-processing steps and data sources
- Knowledge of tools for toxicity prediction

Learning targets and study questions

- Understanding the major concepts for toxicity prediction
  - Discuss the difference between QSAR and Read-Across.
  - Briefly describe what ICEs and ACRs are.
  - Which steps influence the comparability of toxicity predictions.
  - Discuss model validation and the role of splitting.
- Knowledge on pre-processing steps and data sources
  - Describe typical input formats for chemicals.
  - Provide an overview of data that can be useful to predict toxicity

In silico tools for ecotoxicity prediction

Particularly useful when empirical data is lacking



Quantitative Structure Activity Relationships (QSAR)

- Mathematical model
- General form: activity = f(structure)
- Can be applied to estimate biological or physicochemical properties (e.g. persistence, bioaccumulation, distribution)
- May also consider biological predictors when predicting ecotoxicity

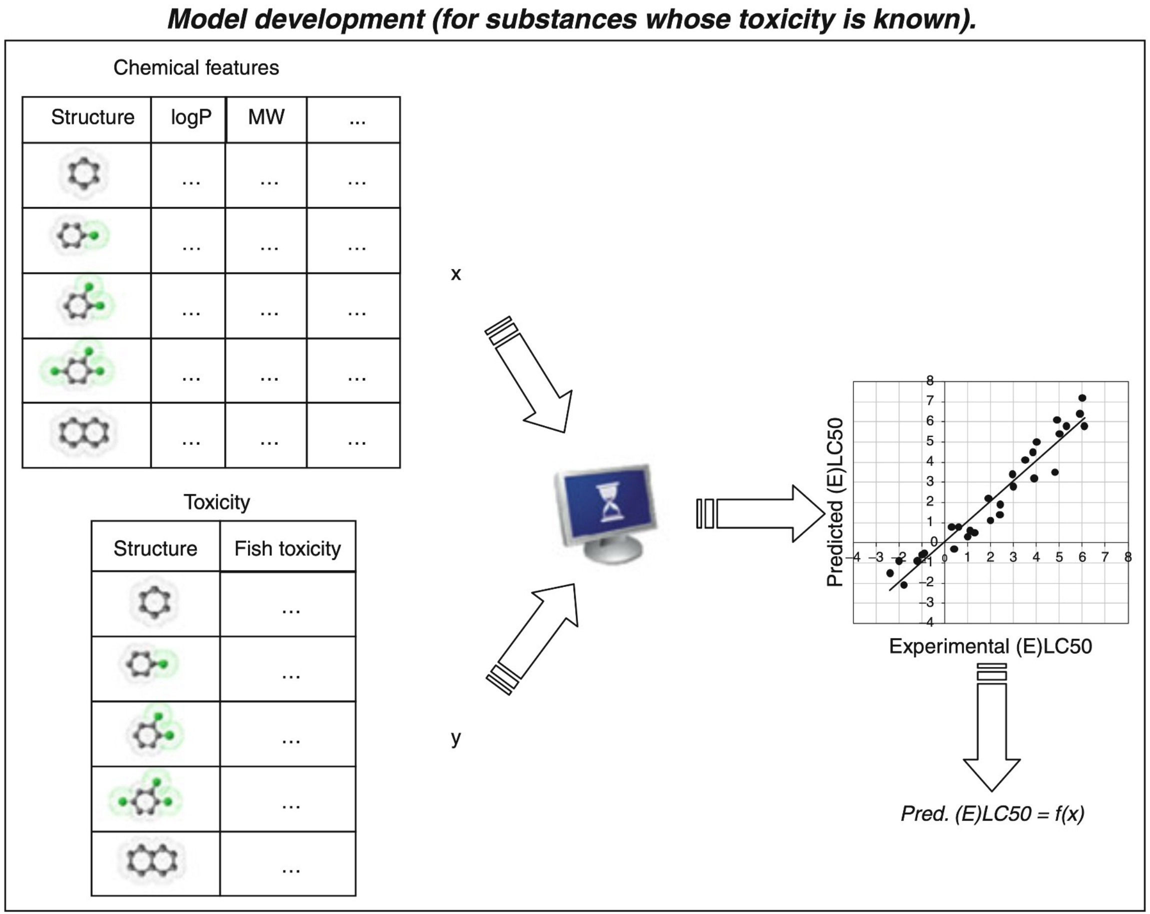
Quantitative Structure Activity Relationships (QSAR)

Example: Model for congeneric chemical

$$\log(1/C) = a(\log P) + b(\epsilon) + c(S) + d$$

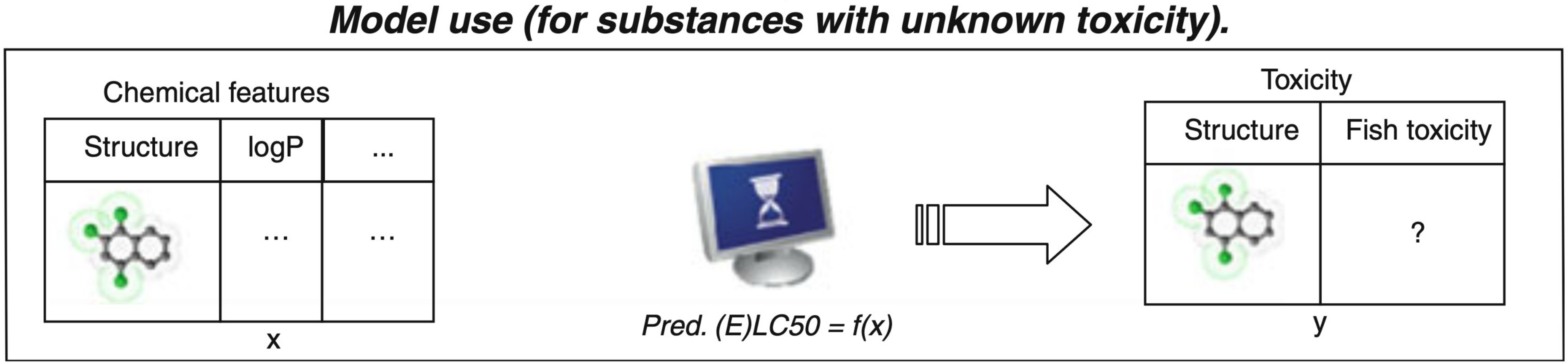
C = effective concentration (e.g. EC50)
P = octanol water partitioning coefficient
ε = Hammett substituent constant
S = Taft's substituent constant (steric)
a, b, c and *d* = model parameters/ functions for variables

General QSAR development



Lomborda et al. 2013

General QSAR development



QSAR model

Prediction can be based on classical statistical (regression) or machine learning model

Lomborda et al. 2013

Input data for *in silico* approaches

Typical input formats (chemicals)

Identifier Type	Value
Chemical name	Caffeine
Molecular formula	$C_8H_{10}N_4O_2$
SMILES	CN1C=NC2=C1C(=O)N(C(=O)N2C)C
InChI	InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3
InChIKey	RYYVLZVUVIJVGH-UHFFFAOYSA-N
CAS number	58-08-2
PubChem CID	2519
IUPAC name	1,3,7-trimethylpurine-2,6-dione

https://pubchem.ncbi.nlm.nih.gov/compound/2519

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For structural representation

For linking to databases

https://pubchem.ncbi.nlm.nih.gov/compound/2519

Input data for *in silico* approaches

Typical chemical input data: Chemical descriptors

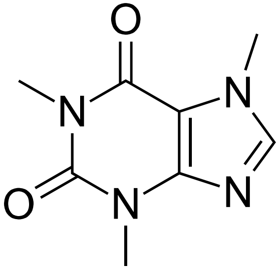
1-D

Scalar physicochemical

- Molecular weight
- Log P
- Water solubility
- …

2-D

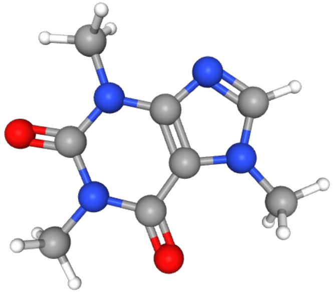
Topological/structural



- No. of atomic fragments (e.g. halogens, rings)
- Connectivity
- Polar surface area
- …

3-D

Spatial/electron distribution



- Steric fields
- Electrostatic maps
- …

https://pubchem.ncbi.nlm.nih.gov/compound/2519

Input data for *in silico* approaches

Typical chemical input data: Biological descriptors

Physiology

- volume-specific somatic maintenance
- maturation threshold for reproduction
- surface-area-specific max. assimilation
- …

Add-my-Pet

https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/species_list.html

Phylogeny

e.g. **ape, taxize** R packages

Traits

- Feeding type
- Life span
- Reproductive rate
- …

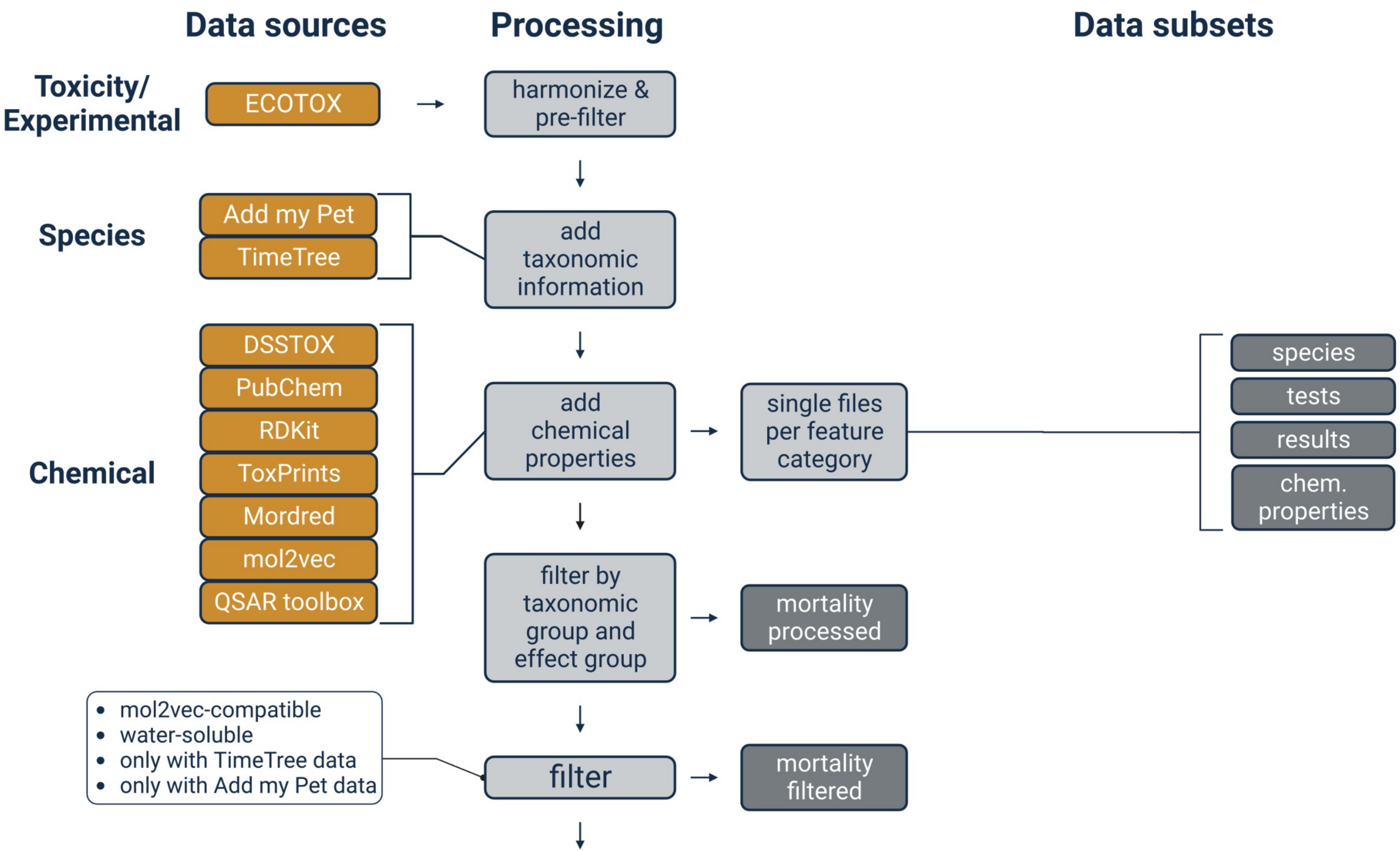
Biogeography

- Ecozone (e.g. geology, altitude)
- Climate
- …

Open Traits Network

https://opentraits.org/datasets.html

Data pre-processing workflow



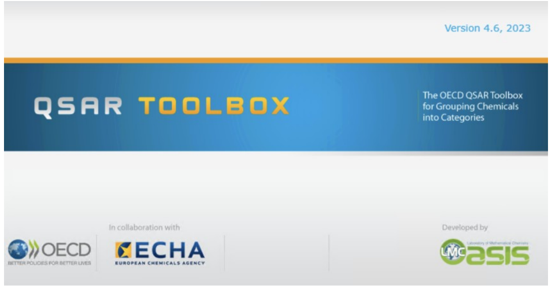
Use in modelling

Schür et al. *Sci. Dat.* 2023

QSAR & RA tools

- QSARs
- Mixtures
- Hazard & Exposure
- TK
- USA EPA QSARs included
- …

https://arnotresearch.com/eas-e-suite/#



- QSARs & RA
- Empirical data
- Contributions by US EPA, EFSA etc.
- …

https://qsartoolbox.org

- QSARs & RA
- Empirical data
- Platform for many models
- …

https://www.vegahub.eu

QSAR & RA tools

CompTox Chemicals Dashboard

Search 1,218,248 Chemicals

- Includes WebTEST
- QSAR & RA
- USA EPA
- …

https://comptox.epa.gov/

OPERA

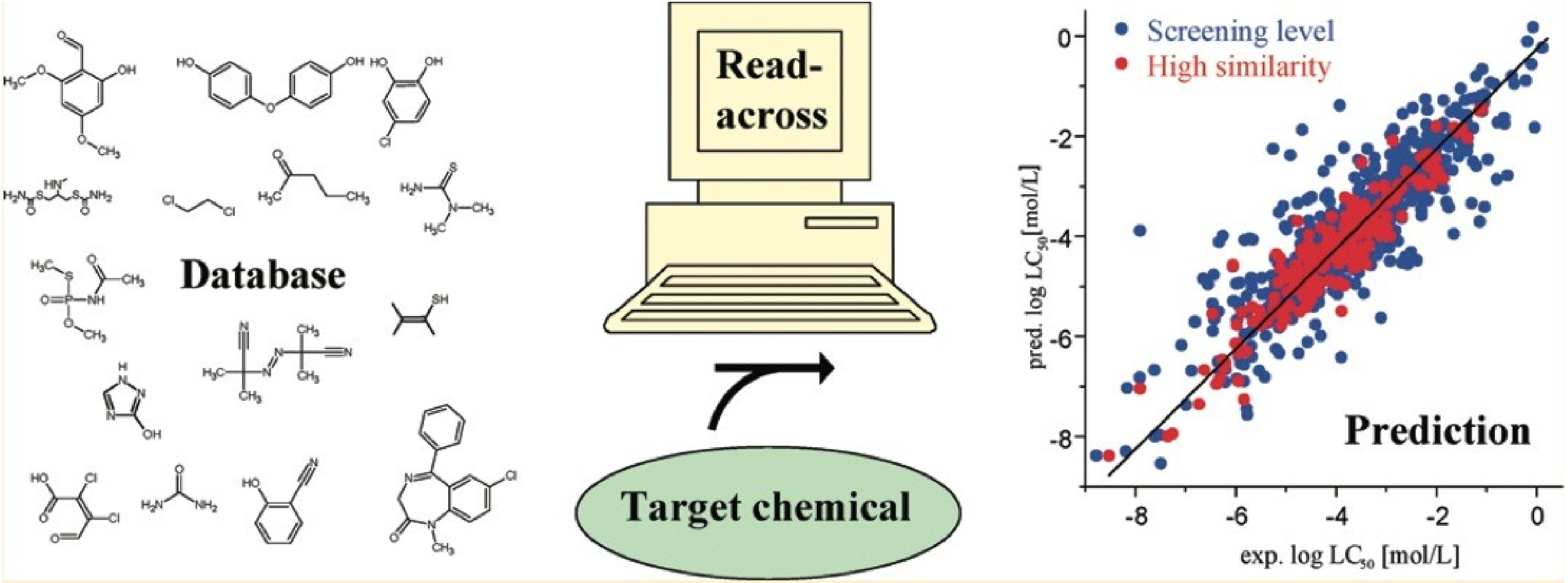
OPERA is a free and open-source/open-data suite of QSAR models providing predictions on physicochemical properties, environmental fate and toxcicity endpoints as well as additional information including applicability domain and accuracy assessment. All models were built on curated data and standardized QSAR-ready chemical structures. OPERA is available in command line and user-friendly graphical interface for Windows and Linux operating systems. It can be installed as a standalone desktop application or embedded in a different tool/workflow.

- Suite of open source QSARs
- Originally developed by US EPA and NIEHS
- …

https://github.com/kmansouri/OPERA

Quantitative Read-Across: Concept

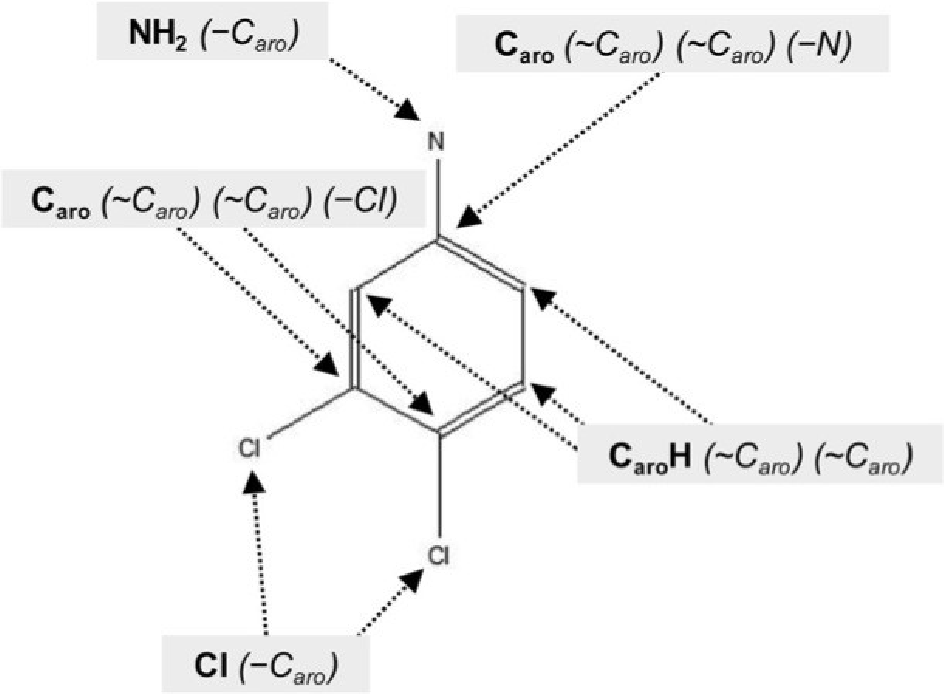
Estimates toxicity based on similarity with compounds of known toxicity



Schüürmann et al. 2011 *Environ. Sci. Technol.* 45: 4616

Quantitative Read-Across: Concept

Similarity based on atomic-centred fragments (ACFs)

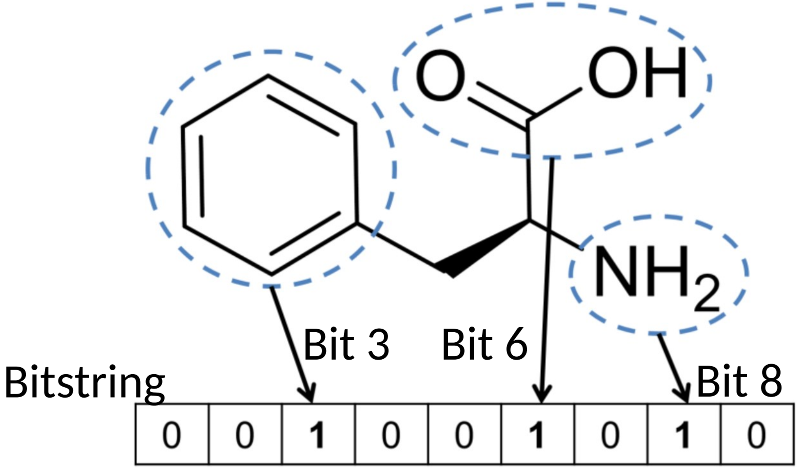


Kühne et al. 2013 *Molecular Informatics* 32 (1): 108

Calculation of Tanimoto similarity

1) Convert molecules to bitstrings

Example:
Molecule A: 0 0 1 0 0 1 0 1 0
Molecule B: 1 1 0 1 0 1 1 0 0



2) Compute similarity measure

		Chemical 1	
		Bit_i present	Bit_i absent
Chemical 2	Bit_i present	a	b
	Bit_i absent	c	d

$$S_{\text{Tanimoto}} = \frac{a}{a+b+c}$$

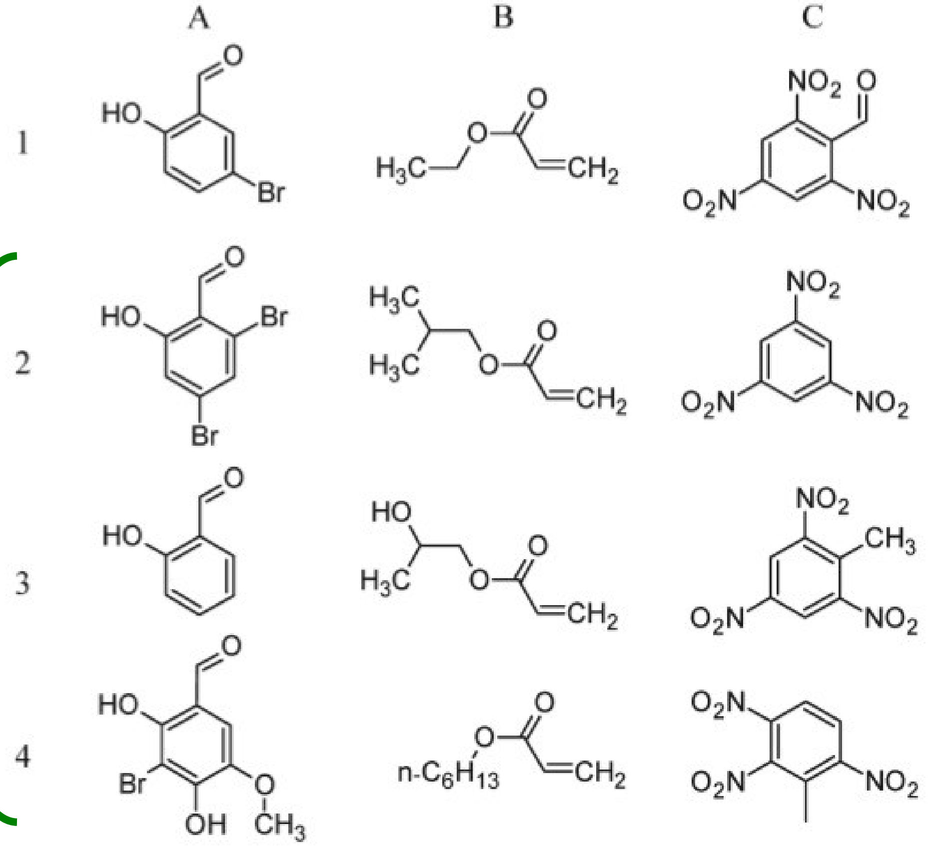
Bajusz et al. 2017

Quantitative Read-Across: Example

Prediction of properties (e.g. physico-chemical, toxicity) weighted by similarity

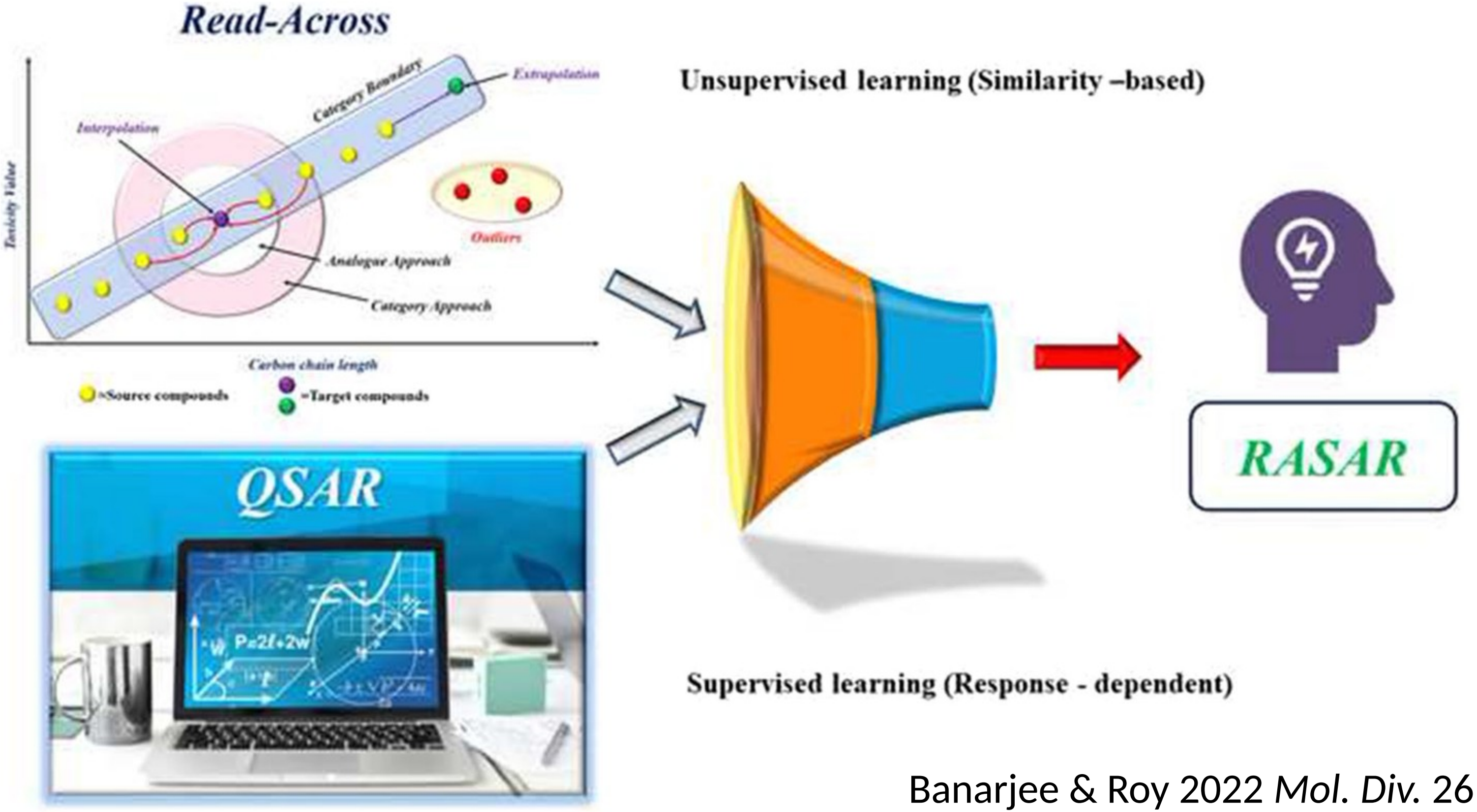
Target compounds

Compounds with known toxicity in decreasing similarity



Schüürmann et al. 2011 *Environ. Sci. Technol.* 45: 4616

qRASAR: Quantitative Read-Across structure-activity relationships



Banarjee & Roy 2022 *Mol. Div.* 26: 2847

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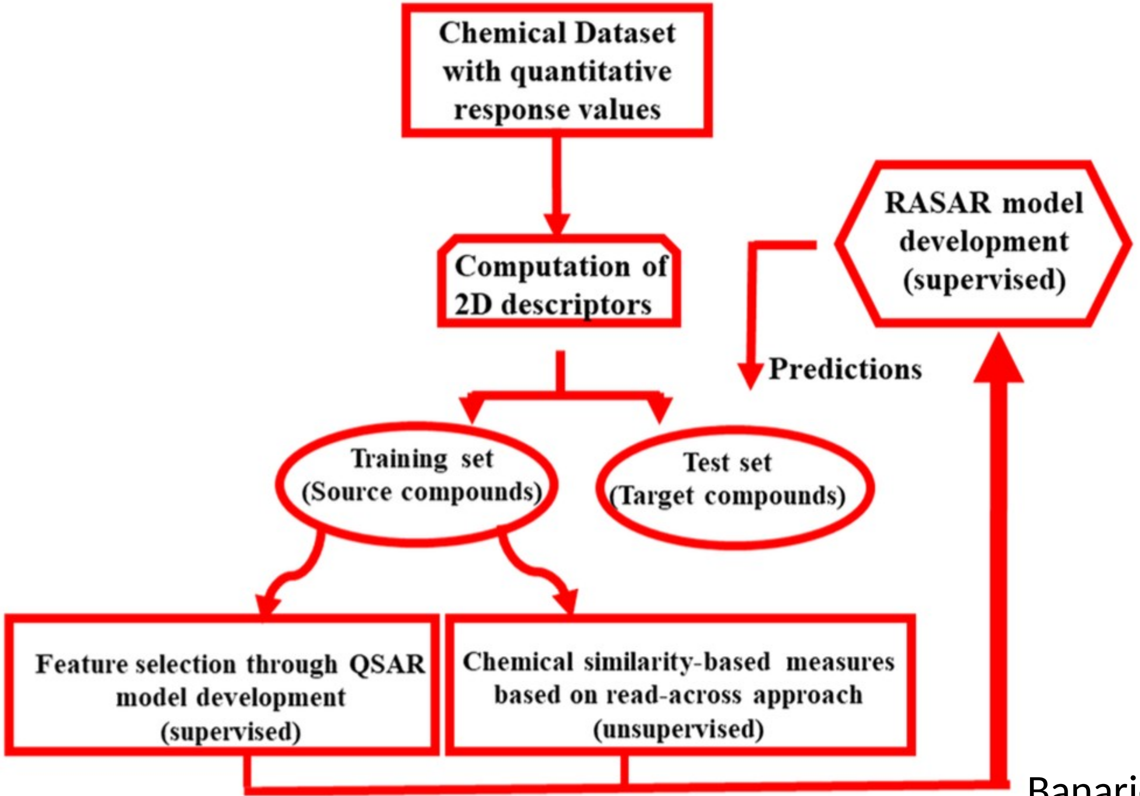


DTC Lab Tools

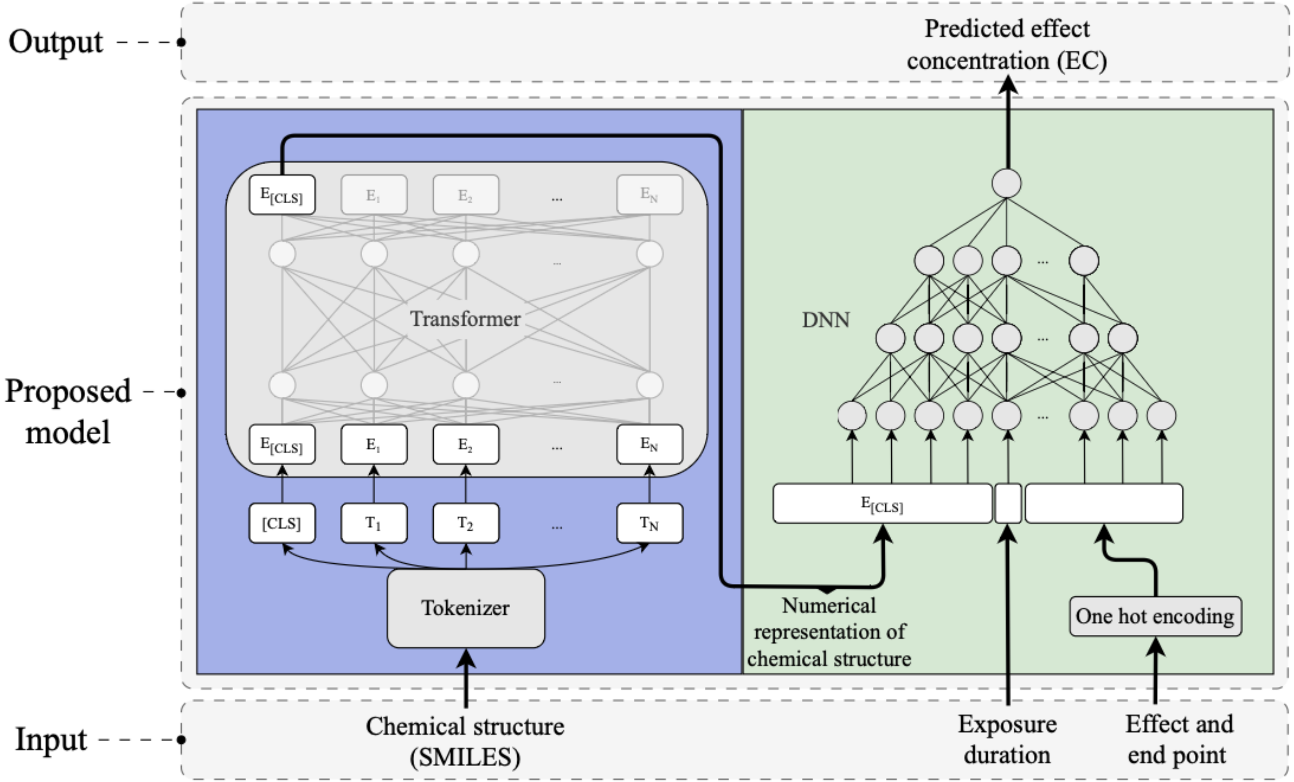
- Prediction tools
- RA
- RASAR
- …

https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home

Banarjee & Roy 2022 *Mol. Div.* 26: 2847

Case study: TRIDENT

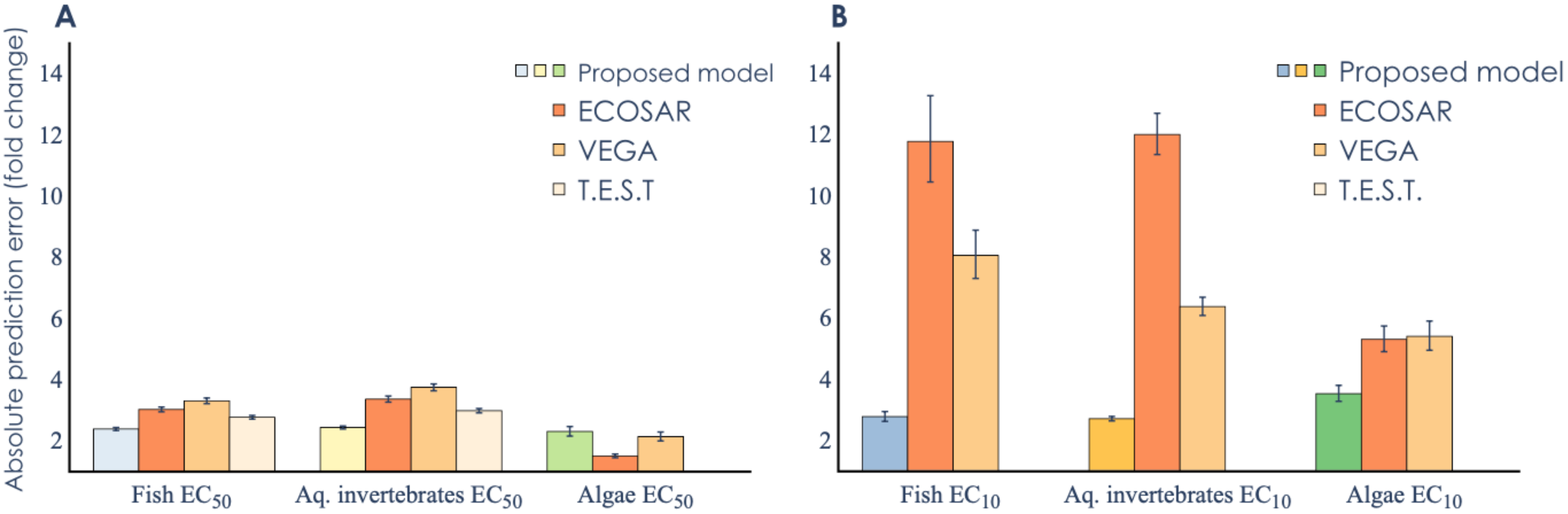
TRIDENT: Based on LLM (BERT) and deep learning



Gustavsson et al. 2024 *Sci. Adv.* 10: eadk6669

Case study: TRIDENT

Generally better performance, in particular for EC_{10}

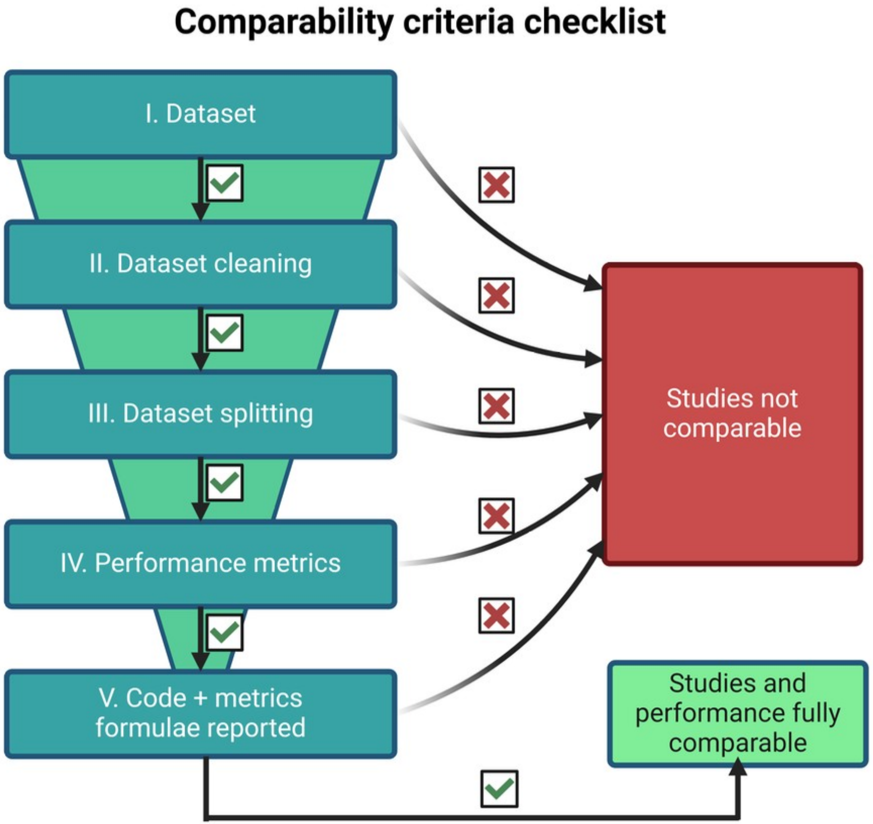


https://ecocait.streamlit.app/

Gustavsson et al. 2024 *Sci. Adv.* 10: eadk6669

Validation of prediction models

What is the best model? Unfortunately, comparability is low



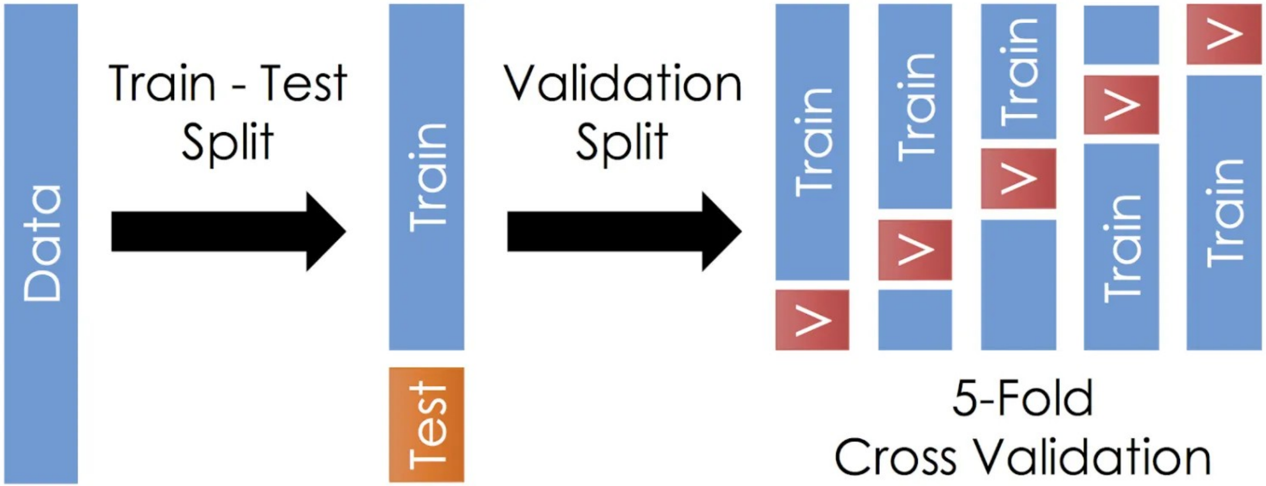
Schür et al. 2025 *Comp. Toxicol.* 35: 100367

Example: Splitting methods

Validation typically done with *n*-fold cross-validation

Typical splits

1) Random
2) By compound
3) By scaffold



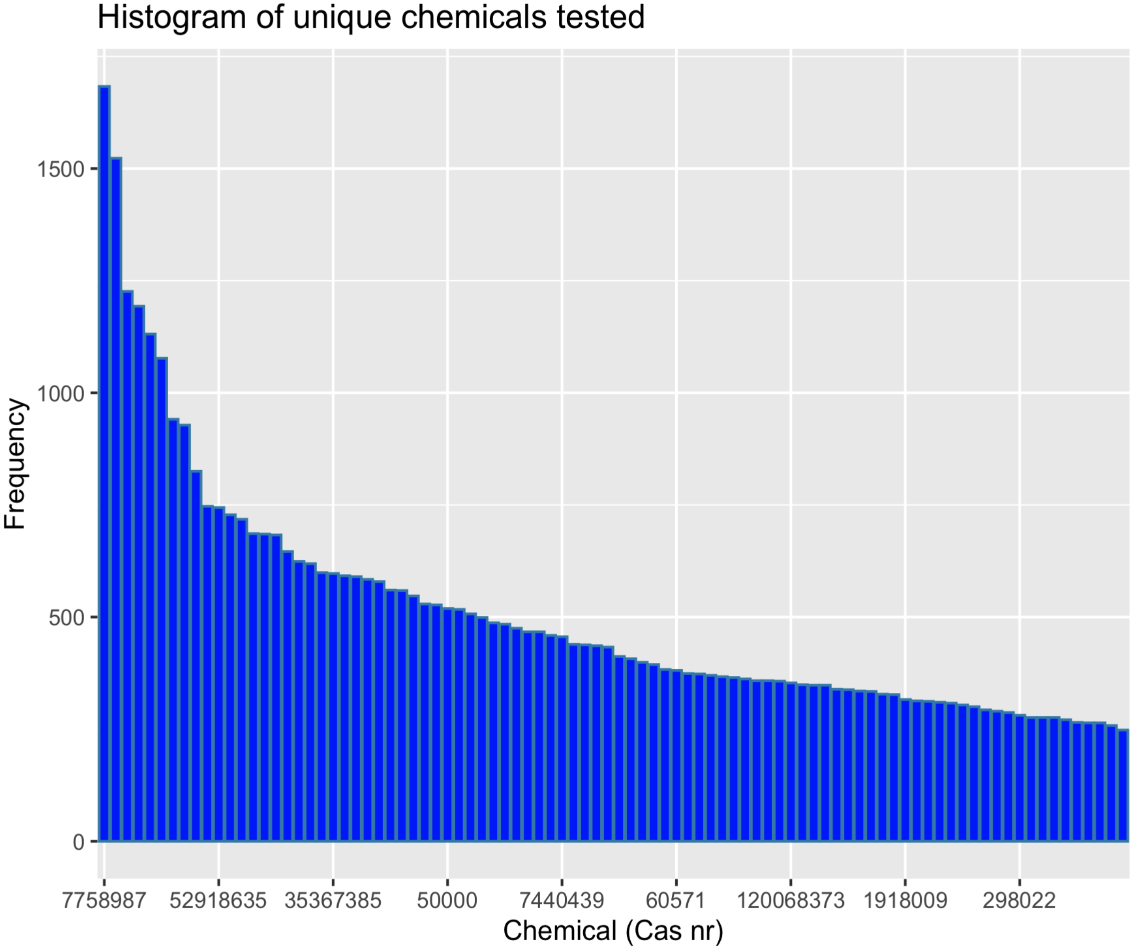
https://medium.com/@hi.martinez/train-test-split-cross-validation-you-b87f662445e1

Example: Splitting methods

Typical splits

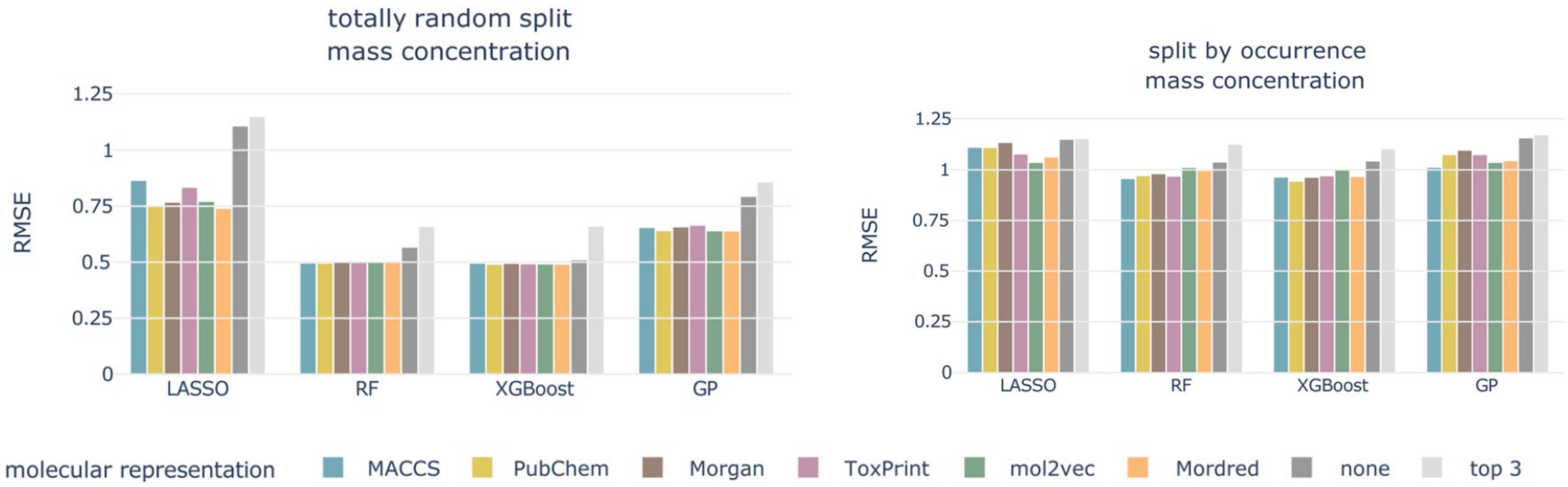
1) Random
2) By compound
3) By scaffold

Random not recommended because high probability of same chemicals in test and training set



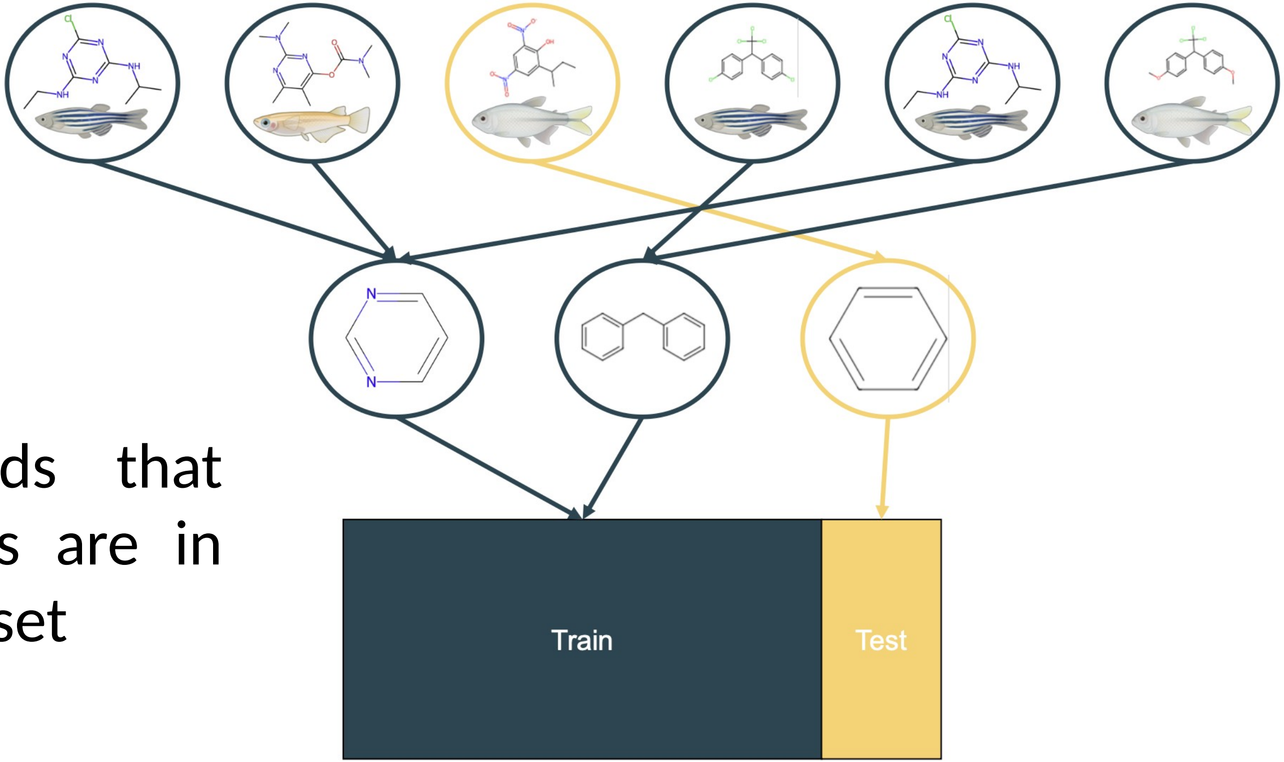
Example: Splitting methods

Much better performance for random split → too optimistic



Gasser et al. 2024 *Environ. Sci. Adv.* 3: 1124

Example: Splitting methods

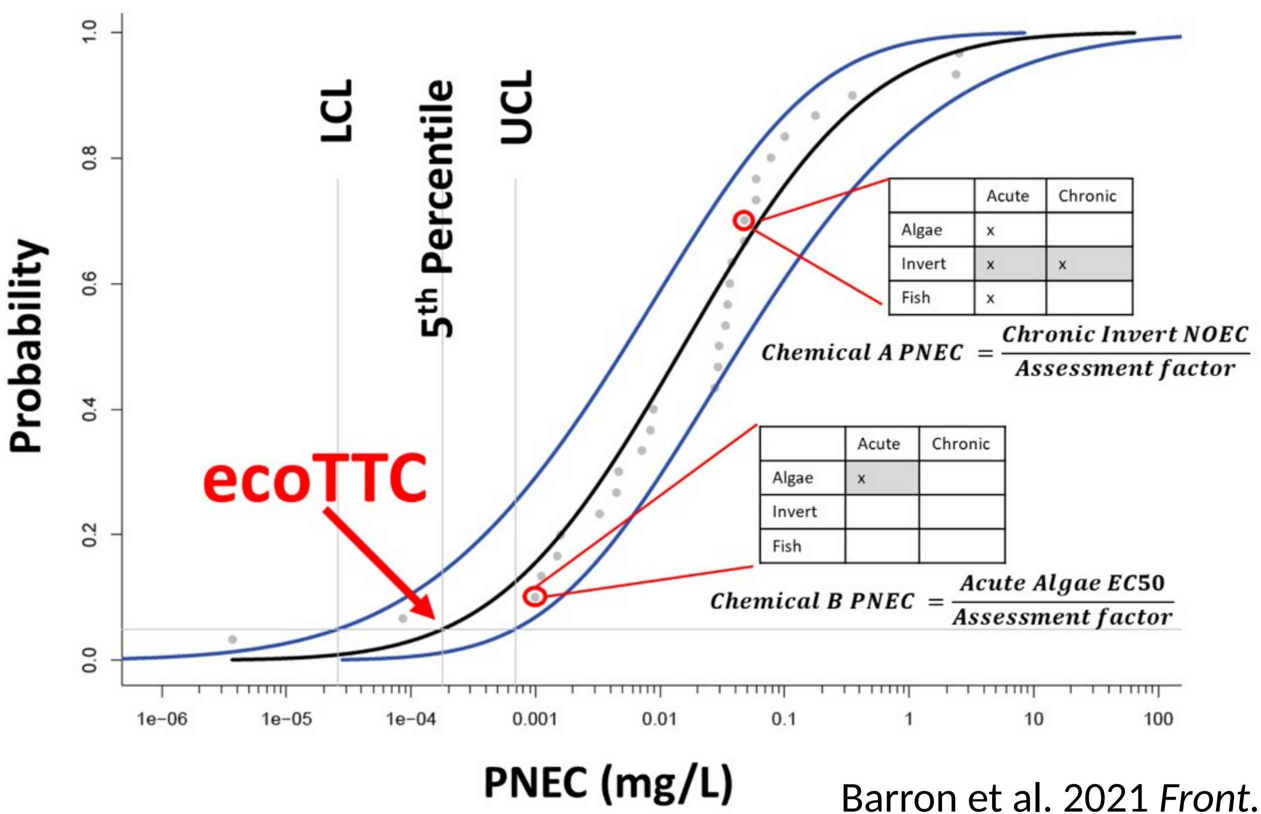


Scaffolding avoids that similar structures are in training and test set

Gasser et al. 2024 *Environ. Sci. Adv.* 3: 1124

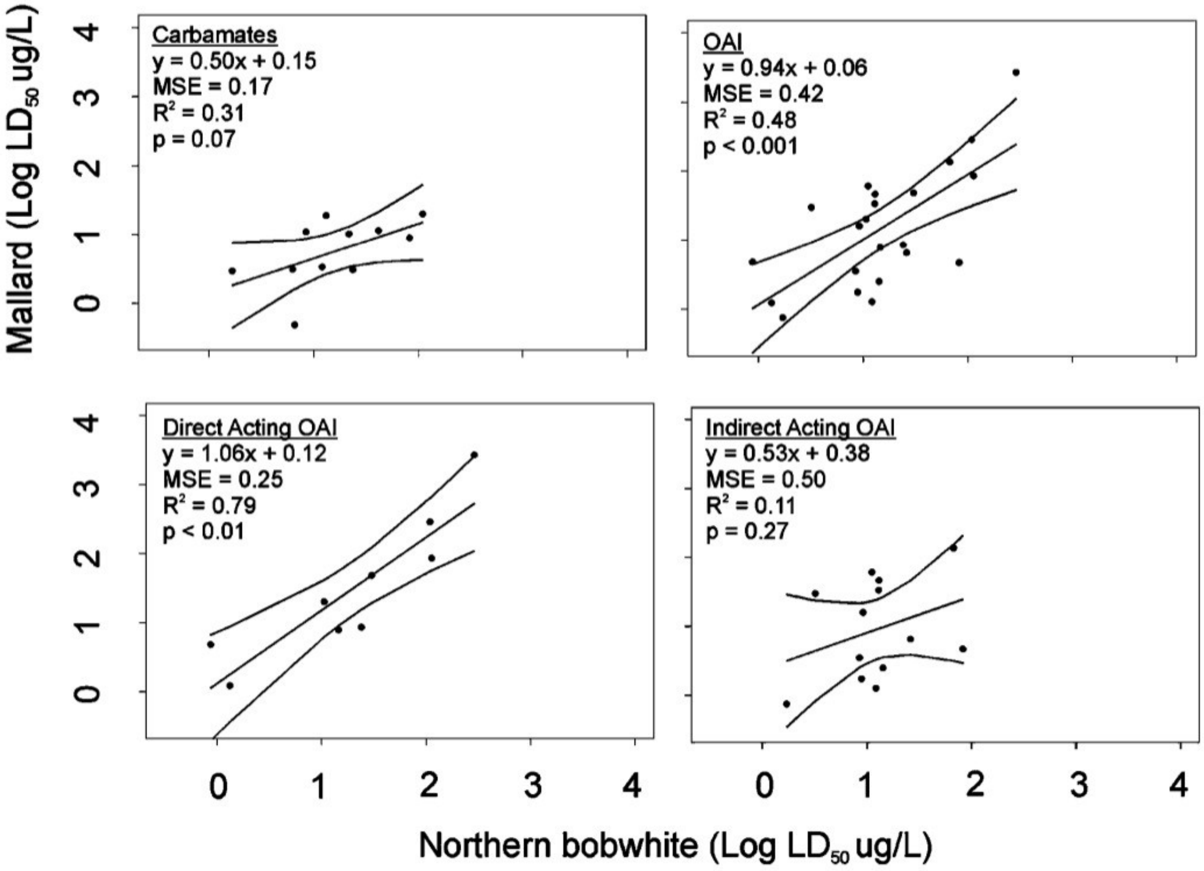
EcoTCC: Ecol. Thresholds of Toxicological Concern

Screening tool for chemicals w/o toxicity data or outside QSAR domain



Barron et al. 2021 *Front. Toxicol.* 3: 640183

Interspecies Correlation Estimate (ICE)



https://www3.epa.gov/webice/

- ICE for aquatic and terrestrial species
- SSDs
- …

https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home

Raimondo 2007 *Environ. Sci. Technol.* 41(16): 5888

Acute-to-Chronic ratio (ACR)

$$ACR = \frac{ECx(acute)}{ECx(chronic)}$$

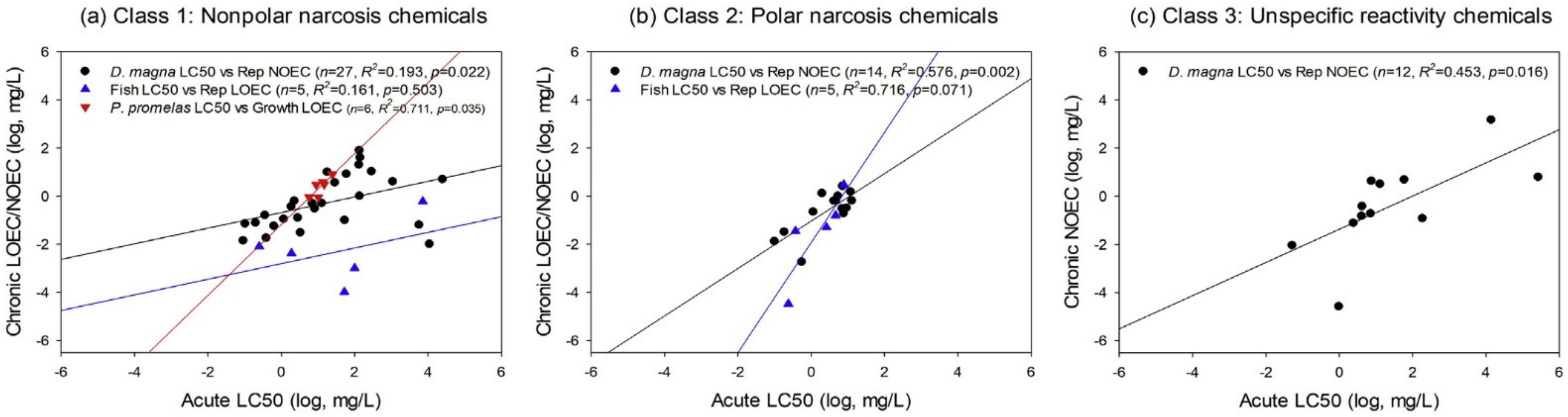
Table 1. Overview of results from acute to chronic ratio (ACR) evaluation for each trophic level

Parameter	Green algae	*Daphnia magna*	Fish
No. of values	104	102	39
No. of substances	102	102	32
50th percentile (median)	5.4	7.0	10.5
90th percentile	33.3	41.5	198.2
Minimum value	1.06	0.6	1.3
Maximum value	4,400	2,222	4,250

Ahlers et al. 2006 *Environ. Toxicol. Chem.* 25 (11): 2937

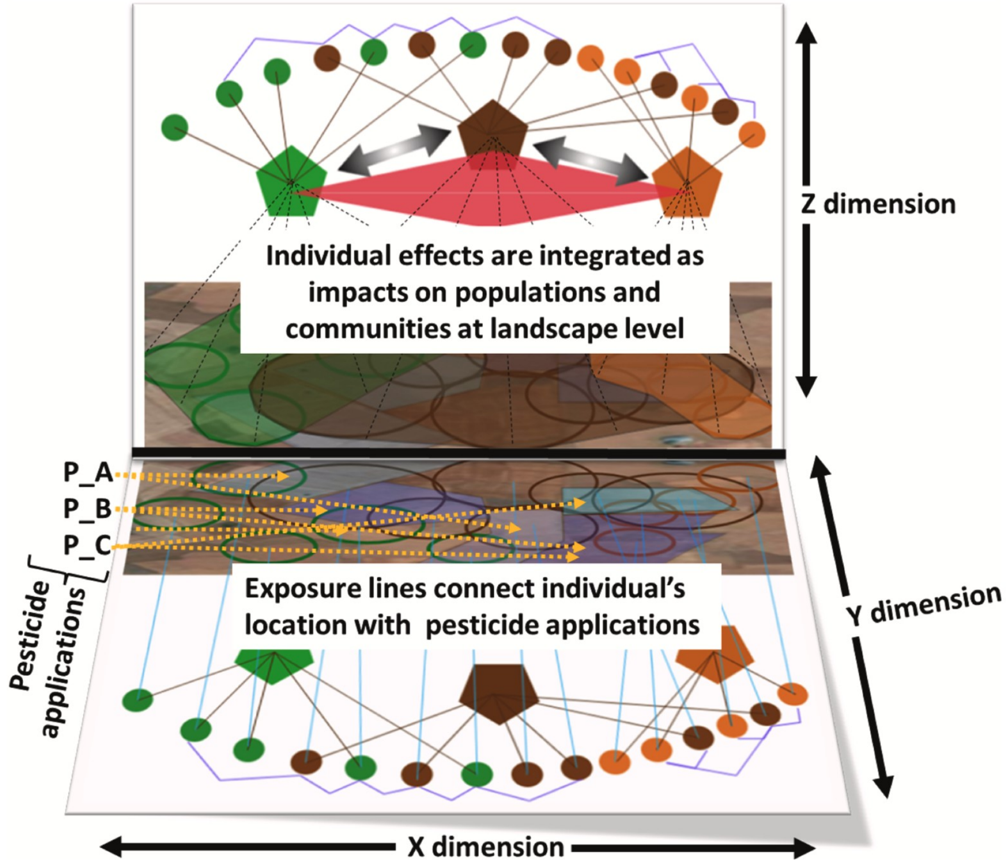
Acute-to-Chronic ratio (ACR)

ACRs dependent on MOA, organism group and test data properties



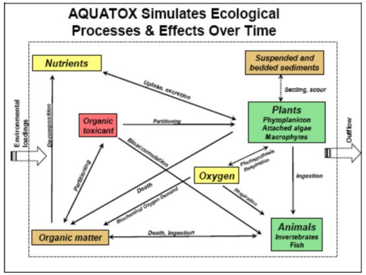
Wang et al. 2020 *Environ. Poll.* 262: 114262

Eco(toxico)logical models



Tarazona et al. 2024 *Environ. Intern.* 191: 108999

Wide range of general and specific eco(toxicological) models available



https://www.epa.gov/hydrowq/aquatox

https://projects.au.dk/almass

Theoretical Ecotoxicology
Session 3: Toxicity prediction

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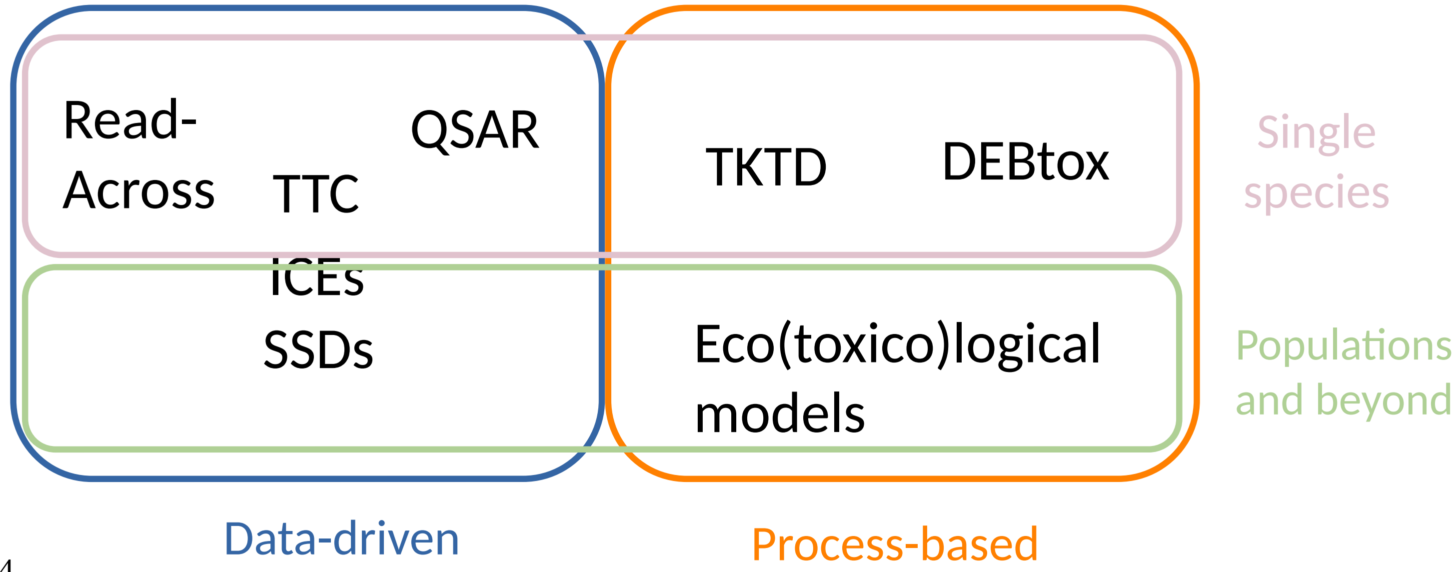

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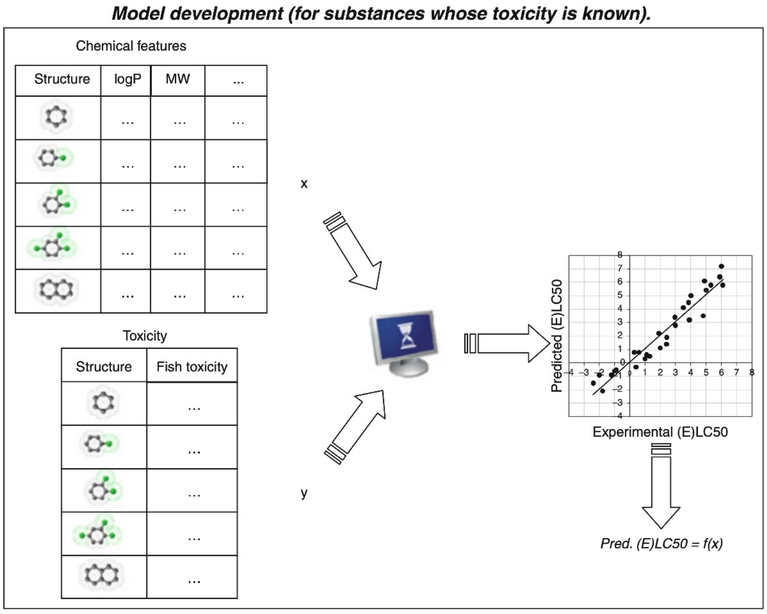
6

Congeneric chemicals share the same core chemical structure, but have different substituents. They typically have similar physicochemical and biological properties.
Examples: PCBs share same biphenyl structure, PAHs are characterised by multiple fused aromatic rings and PFAS share a perfluorinated carbon chain.

General QSAR development



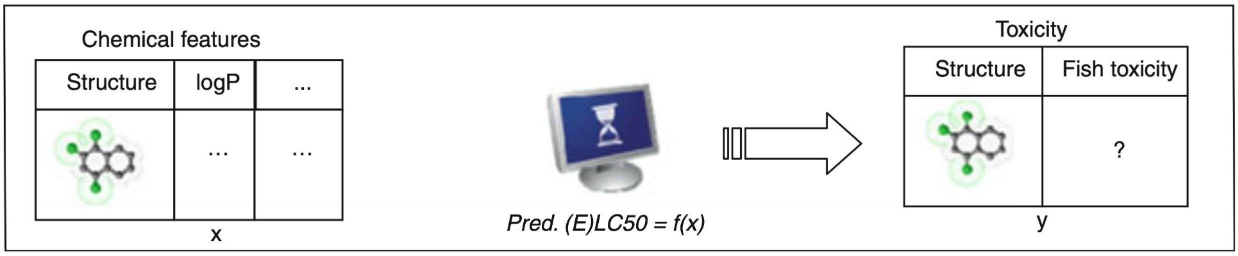
7

Lomborda et al. 2013

Lombardo A., Schifanella O., Roncaglioni A., Benfenati E. 2010: Quantitative Structure-Activity Relationship (QSAR) in Ecotoxicology. In: Férard J., Blaise C. (Ed.) Encyclopedia of Aquatic Ecotoxicology: SpringerReference (www.springerreference.com). Springer-Verlag Berlin Heidelberg, 2013. DOI: 10.1007/SpringerReference_380001

General QSAR development

Model use (for substances with unknown toxicity).



QSAR model

Prediction can be based on classical statistical (regression) or machine learning model

Lomborda et al. 2013

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For structural representation

For linking to databases

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Canonical SMILES (Simplified Molecular Input Line Entry System), but not standard SMILES, and the InChI (International Chemical Identifier (IUPAC-based)) are unique and capture stereochemistry. The InChIkey is a hashed version of InChI and is more suitable for indexing and searching, but cannot be reversed to structure.

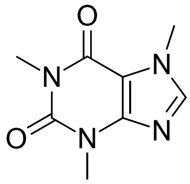
Input data for *in silico* approaches

Typical chemical input data: Chemical descriptors

1-D
Scalar physicochemical

- Molecular weight
- Log P
- Water solubility
- …

2-D
Topological/structural



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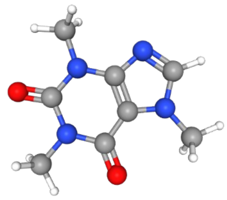
- Connectivity
- Polar surface area
- …

3-D
Spatial/electron distribution



- Steric fields
- Electrostatic maps
- …

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Many sources provide chemical descriptors (e.g. Mordred, Morgan, mol2vec), for an overview see: Schür, C., Gasser, L., Perez-Cruz, F., Schirmer, K., & Baity-Jesi, M. (2023). A benchmark dataset for machine learning in ecotoxicology. Scientific Data, 10(1), 718. https://doi.org/10.1038/s41597-023-02612-2

Input data for *in silico* approaches

Typical chemical input data: Biological descriptors

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- volume-specific somatic maintenance
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- …

Add-my-Pet

https://www.bio.vu.nl/thb/deb/deblab/add_my_pet/species_list.html

Phylogeny

e.g. **ape, taxize** R packages

Traits

- Feeding type
- Life span
- Reproductive rate
- …

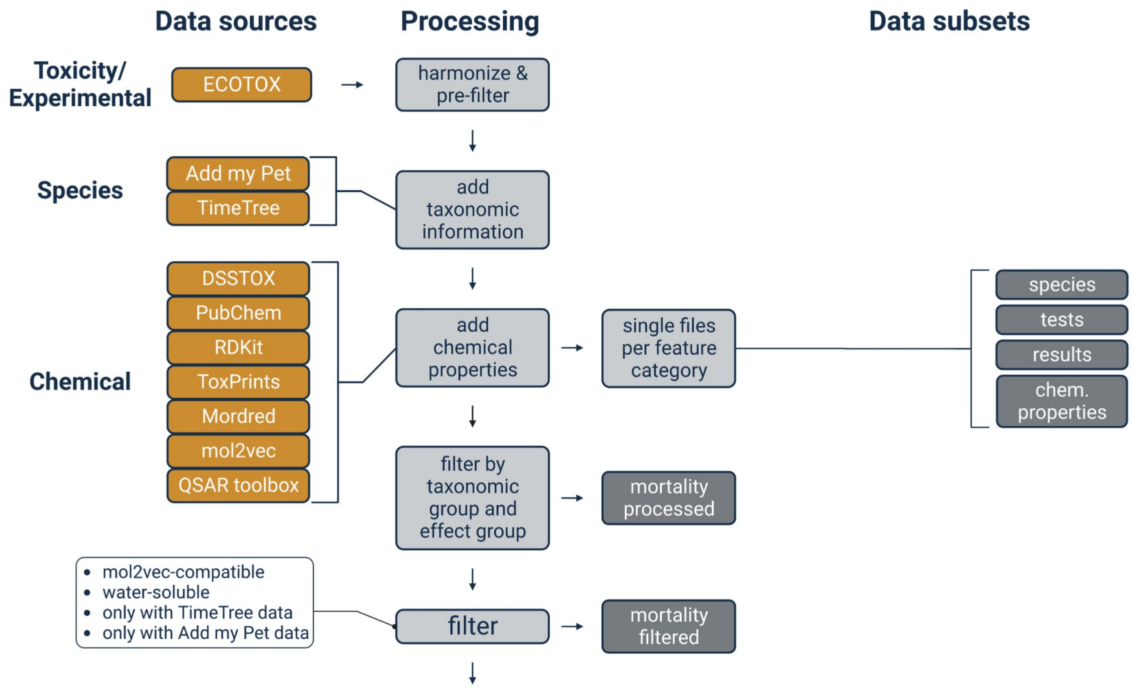
Biogeography

- Ecozone (e.g. geology, altitude)
- Climate
- …

Open Traits Network

https://opentraits.org/datasets.html

Data pre-processing workflow



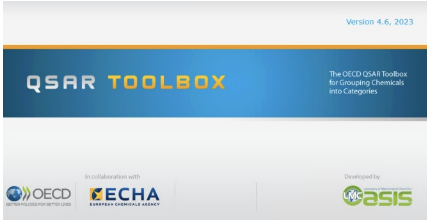
Use in modelling

Schür et al. *Sci. Dat.* 2023

QSAR & RA tools

- QSARs
- Mixtures
- Hazard & Exposure
- TK
- USA EPA QSARs included
- …

https://arnotresearch.com/eas-e-suite/#



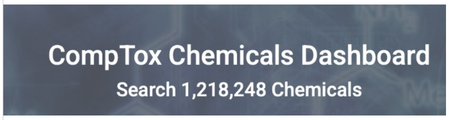
- QSARs & RA
- Empirical data
- Contributions by US EPA, EFSA etc.
- …

https://qsartoolbox.org

- QSARs & RA
- Empirical data
- Platform for many models
- …

https://www.vegahub.eu

QSAR & RA tools



- Includes WebTEST
- QSAR & RA
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https://comptox.epa.gov/

OPERA

OPERA is a free and open-source/open-data suite of QSAR models providing predictions on physicochemical properties, environmental fate and toxicity endpoints as well as additional information including applicability domain and accuracy assessment. All models were built on curated data and standardized QSAR-ready chemical structures. OPERA is available in command line and user-friendly graphical interface for Windows and Linux operating systems. It can be installed as a standalone desktop application or embedded in a different tool/workflow.

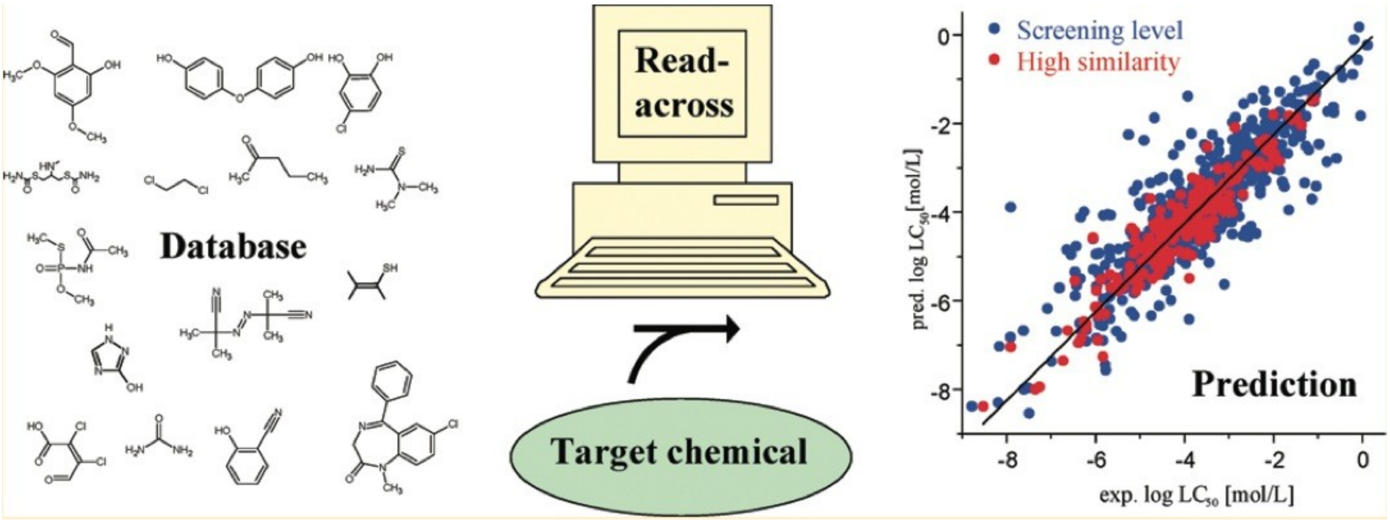
- Suite of open source QSARs
- Originally developed by US EPA and NIEHS
- …

https://github.com/kmansouri/OPERA

You can download a local version of the Toxicity Estimation Software Tool (TEST) under: https://www.epa.gov/comptox-tools/toxicity-estimation-software-tool-test

Quantitative Read-Across: Concept

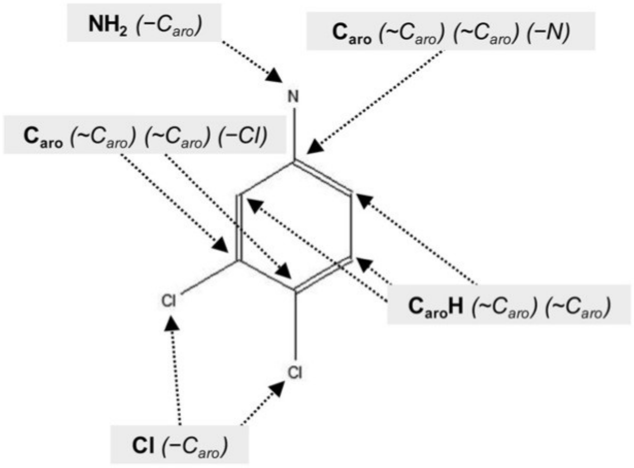
Estimates toxicity based on similarity with compounds of known toxicity



Schüürmann et al. 2011 *Environ. Sci. Technol.* 45: 4616

Quantitative Read-Across: Concept

Similarity based on atomic-centred fragments (ACFs)



17

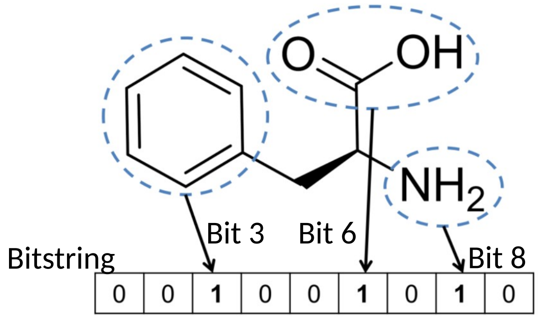
Kühne et al. 2013 *Molecular Informatics* 32 (1): 108

The chemical is decomposed into fragments, which are used to compute the similarity between compounds.

Calculation of Tanimoto similarity

1) Convert molecules to bitstrings

Example:
Molecule A: 0 0 1 0 0 1 0 1 0
Molecule B: 1 1 0 1 0 1 1 0 0



2) Compute similarity measure

		Chemical 1	
		Bit_i present	Bit_i absent
Chemical 2	Bit_i present	a	b
	Bit_i absent	c	d

$$S_{\text{Tanimoto}} = \frac{a}{a+b+c}$$

18

Bajusz et al. 2017

The Tanimoto similarity is identical to the Jaccard similarity in ecology.

a = 1, b = 2, c = 4

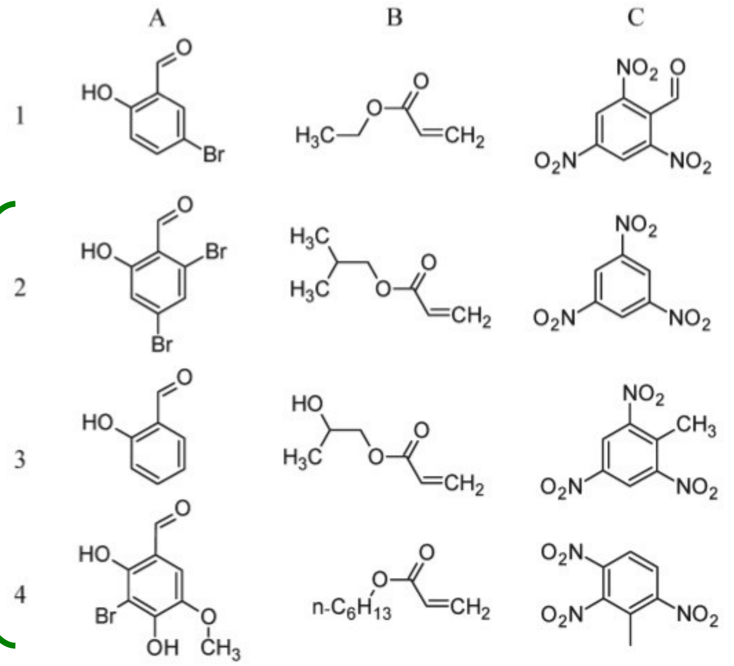
Bajusz, D., Rácz, A., & Héberger, K. (2017). Chemical Data Formats, Fingerprints, and Other Molecular Descriptions for Database Analysis and Searching. In *Comprehensive Medicinal Chemistry III* (pp. 329–378). Elsevier. https://doi.org/10.1016/B978-0-12-409547-2.12345-5

Quantitative Read-Across: Example

Prediction of properties (e.g. physico-chemical, toxicity) weighted by similarity

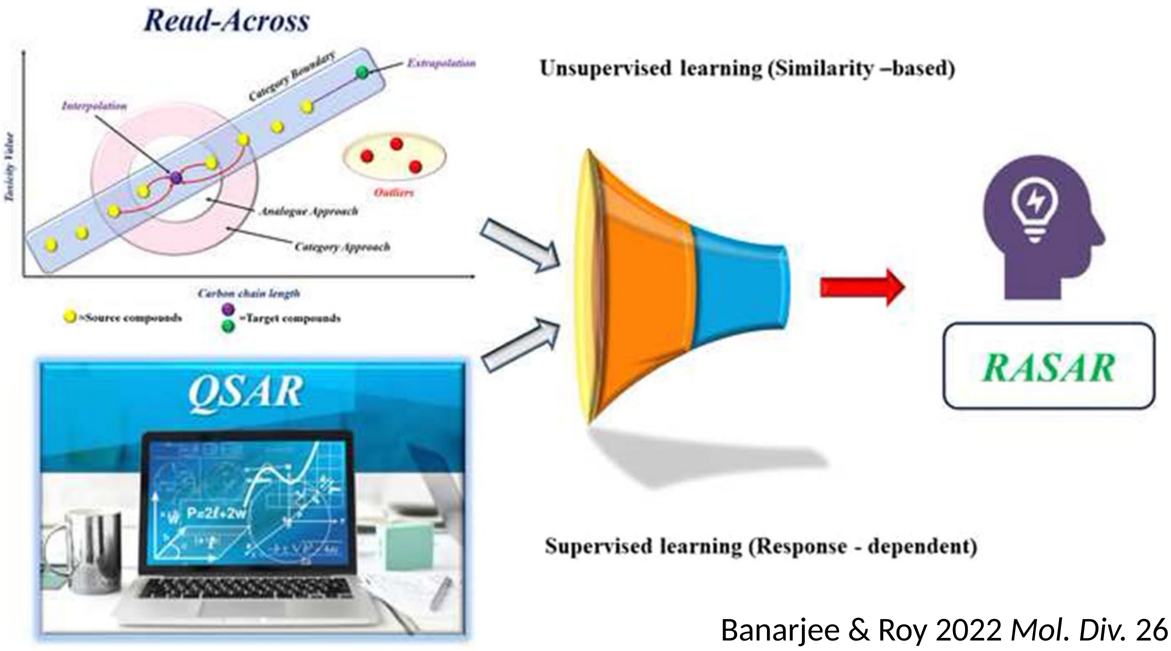
Target compounds

Compounds with known toxicity in decreasing similarity



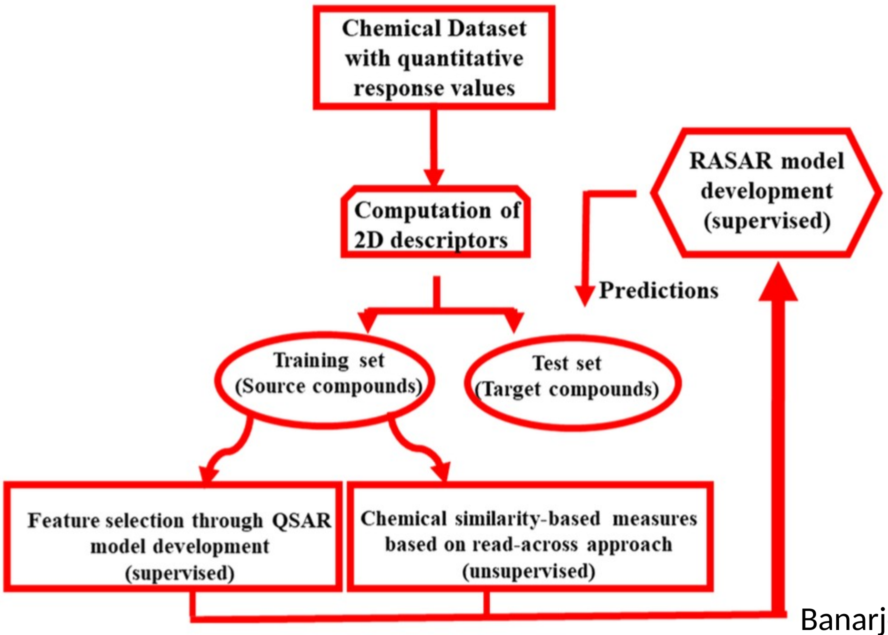
Schüürmann et al. 2011 *Environ. Sci. Technol.* 45: 4616

qRASAR: Quantitative Read-Across structure-activity relationships



Banarjee & Roy 2022 *Mol. Div.* 26: 2847

qRASAR: Quantitative Read-Across structure-activity relationships



DTC Lab Tools

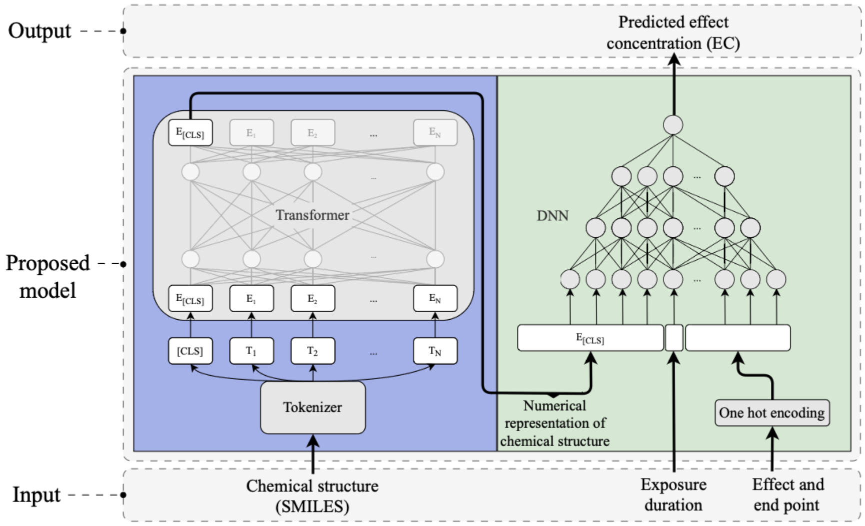
- Prediction tools
- RA
- RASAR
- …

https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home

Banarjee & Roy 2022 *Mol. Div.* 26: 2847

Case study: TRIDENT

TRIDENT: Based on LLM (BERT) and deep learning



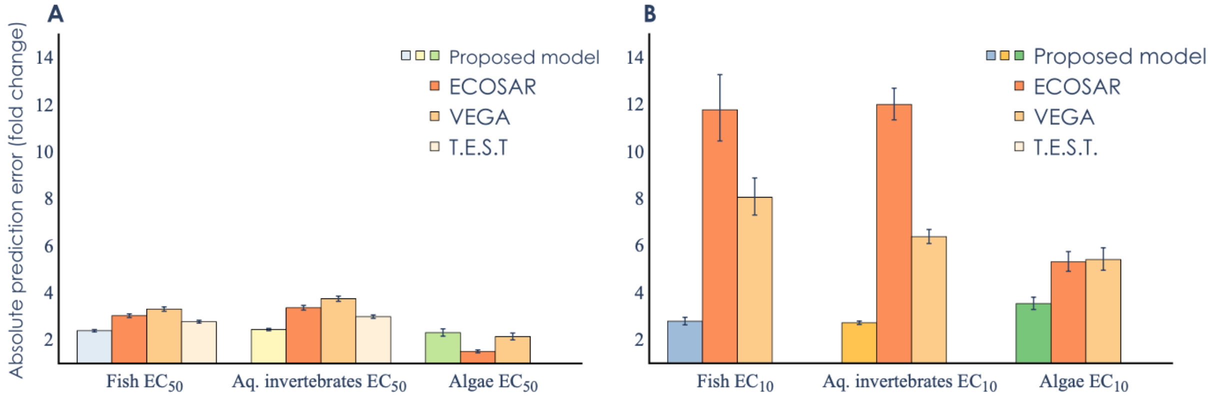
22

Gustavsson et al. 2024 *Sci. Adv.* 10: eadk6669

The background of the language processing is explained here along with the computer code: https://deepchem.io/tutorials/transfer-learning-with-chemberta-transformers/

Case study: TRIDENT

Generally better performance, in particular for EC_{10}



TRIDENT
Predicting chemical ecotoxicity using AI

https://ecocait.streamlit.app/

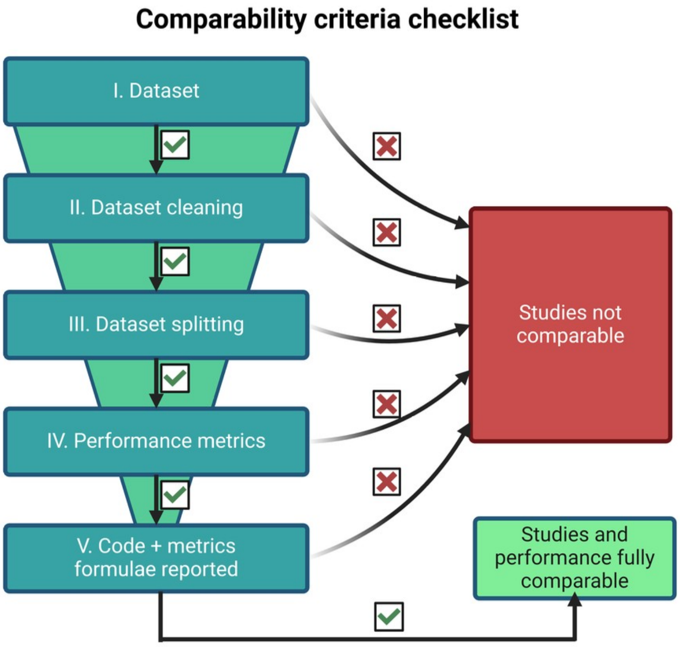
23

Gustavsson et al. 2024 *Sci. Adv.* 10: eadk6669

For another application see the following study that incorporated a wider range of biological information:
Zubrod, J. P., Galic, N., Vaugeois, M., & Dreier, D. A. (2024). Bio-QSARs 2.0: Unlocking a new level of predictive power for machine learning-based ecotoxicity predictions by exploiting chemical and biological information. Environment International, 186, 108607. https://doi.org/10.1016/j.envint.2024.108607

Validation of prediction models

What is the best model? Unfortunately, comparability is low



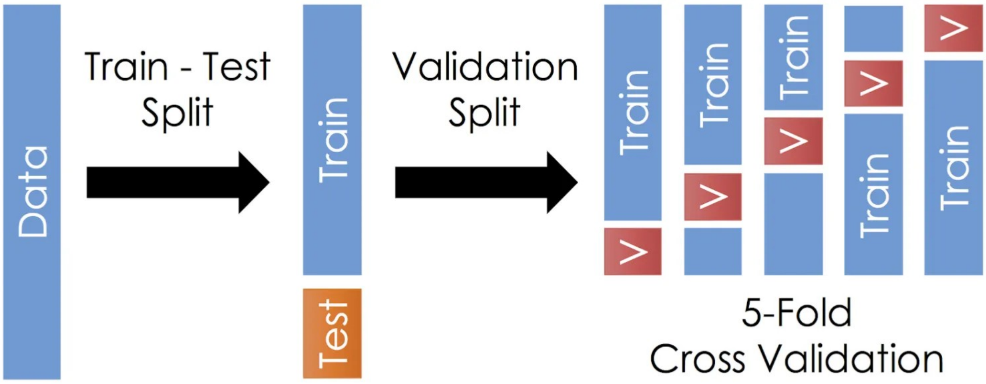
Schür et al. 2025 *Comp. Toxicol.* 35: 100367

Example: Splitting methods

Validation typically done with *n*-fold cross-validation

Typical splits

1) Random
2) By compound
3) By scaffold



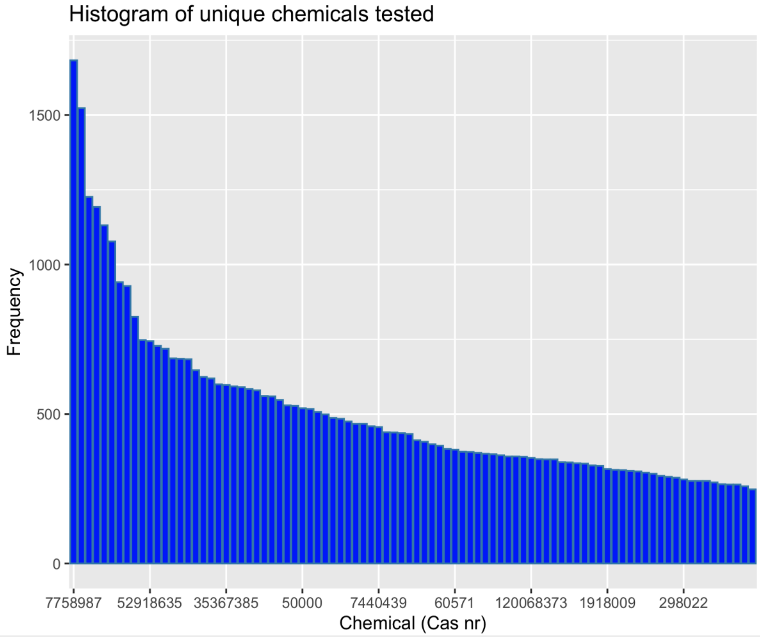
https://medium.com/@hi.martinez/train-test-split-cross-validation-you-b87f662445e1

Example: Splitting methods

Typical splits

1) Random
2) By compound
3) By scaffold

Random not recommended because high probability of same chemicals in test and training set

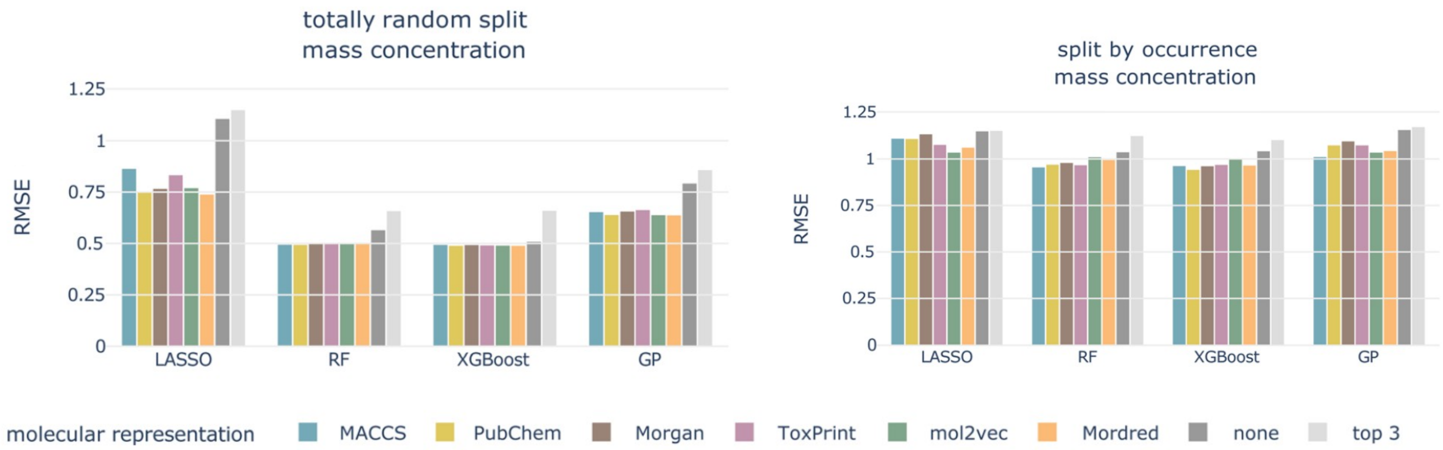


26

Own calculations of the number of tests per unique chemical in the EPA Ecotox dataset in March 2025.

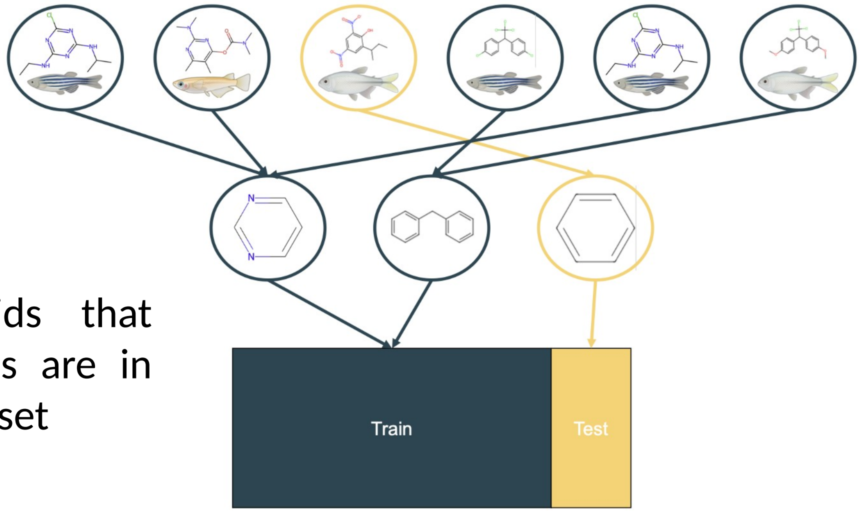
Example: Splitting methods

Much better performance for random split → too optimistic



Gasser et al. 2024 *Environ. Sci. Adv.* 3: 1124

Example: Splitting methods



Scaffolding avoids that similar structures are in training and test set

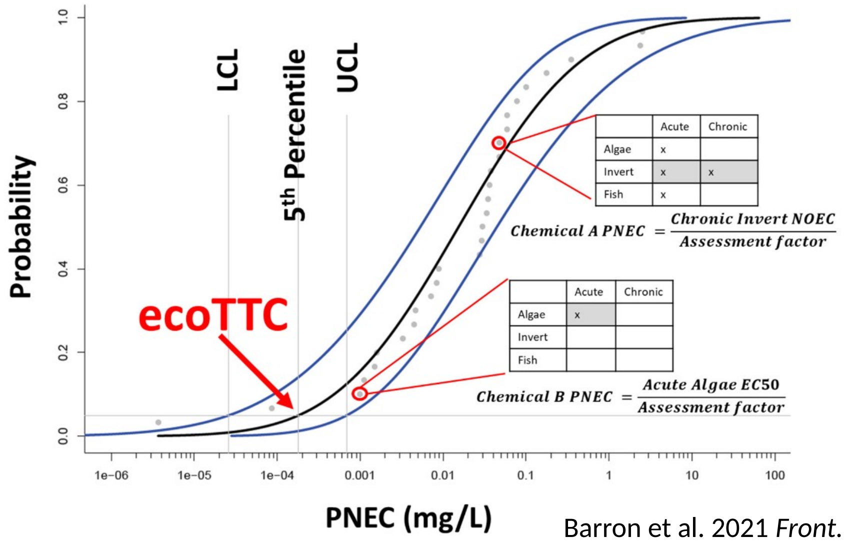
28

Gasser et al. 2024 *Environ. Sci. Adv.* 3: 1124

In case of prediction of compounds of a similar chemical class, scaffolding would neither be applicable (nor required).

EcoTCC: Ecol. Thresholds of Toxicological Concern

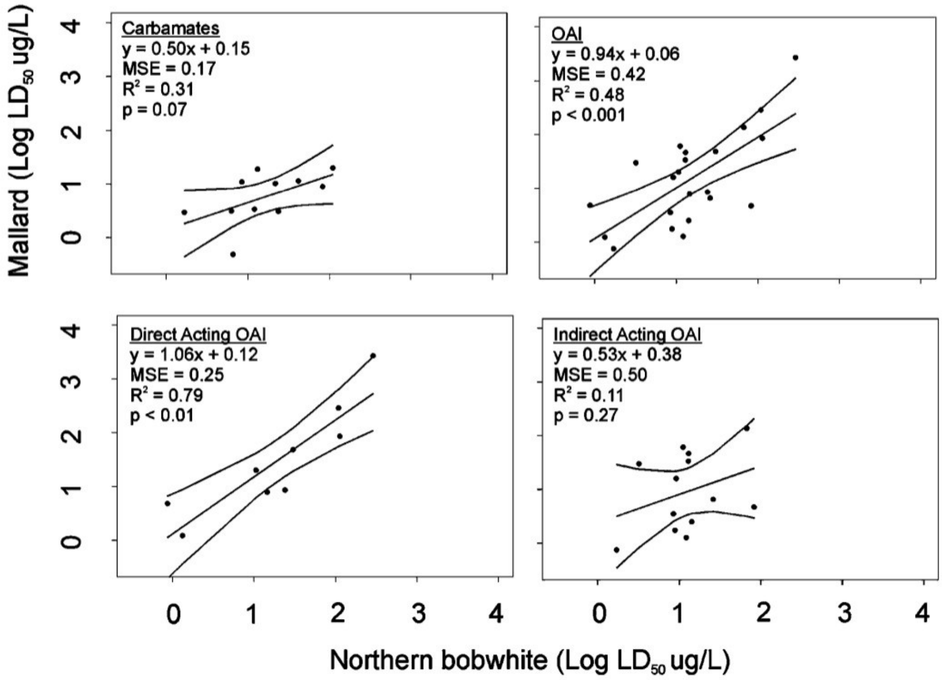
Screening tool for chemicals w/o toxicity data or outside QSAR domain



Barron et al. 2021 *Front. Toxicol.* 3: 640183

A related tool is avaible in the EnviroTox database: https://envirotoxdatabase.org

Interspecies Correlation Estimate (ICE)



https://www3.epa.gov/webice/

- ICE for aquatic and terrestrial species
- SSDs
- …

https://sites.google.com/jadavpuruniversity.in/dtc-lab-software/home

Raimondo 2007 *Environ. Sci. Technol.* 41(16): 5888

Acute-to-Chronic ratio (ACR)

$$ACR = \frac{ECx(acute)}{ECx(chronic)}$$

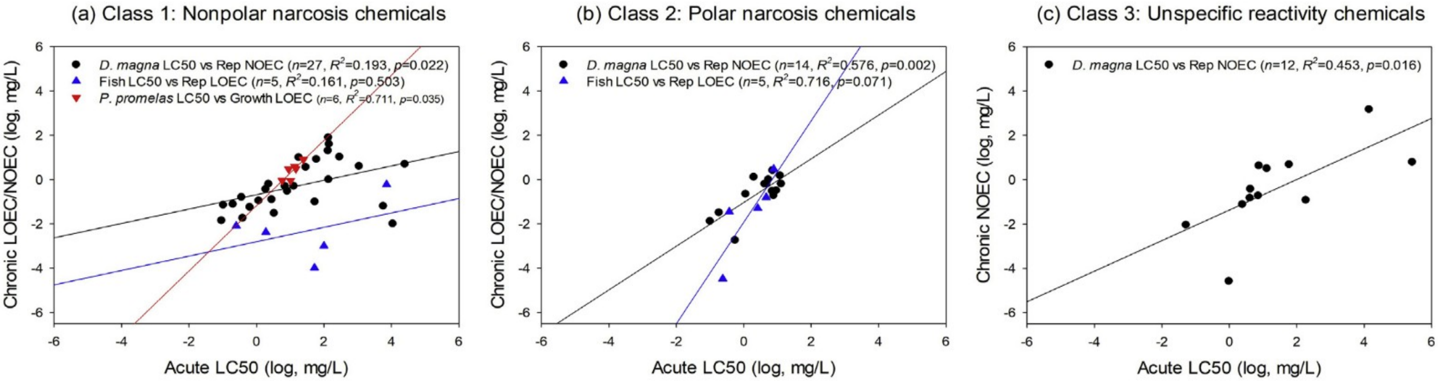
Table 1. Overview of results from acute to chronic ratio (ACR) evaluation for each trophic level

Parameter	Green algae	*Daphnia magna*	Fish
No. of values	104	102	39
No. of substances	102	102	32
50th percentile (median)	5.4	7.0	10.5
90th percentile	33.3	41.5	198.2
Minimum value	1.06	0.6	1.3
Maximum value	4,400	2,222	4,250

Ahlers et al. 2006 *Environ. Toxicol. Chem.* 25 (11): 2937

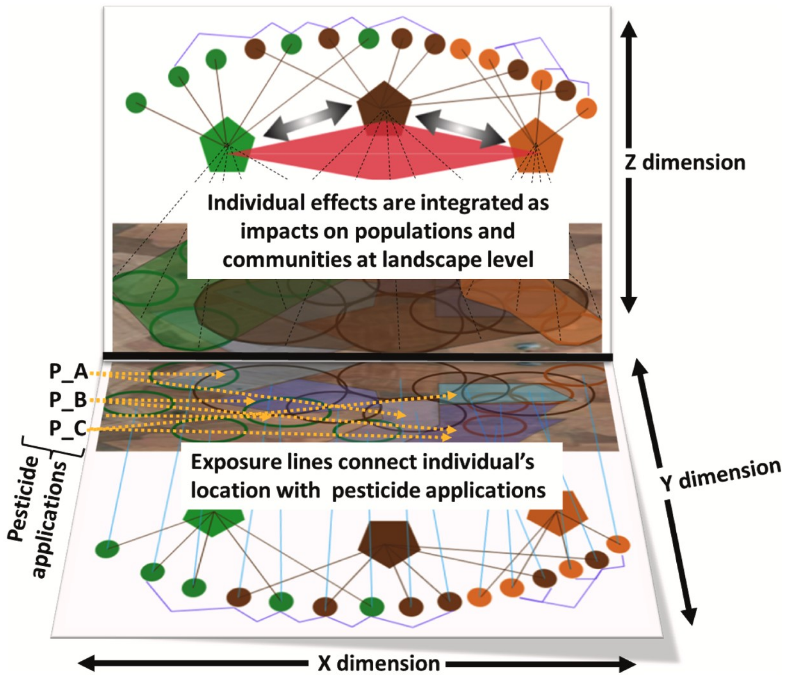
Acute-to-Chronic ratio (ACR)

ACRs dependent on MOA, organism group and test data properties

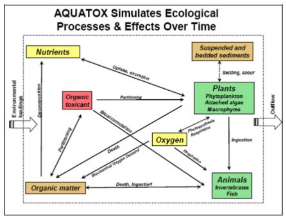


Wang et al. 2020 *Environ. Poll.* 262: 114262

Eco(toxico)logical models



Wide range of general and specific eco(toxicological) models available



https://www.epa.gov/hydrowq/aquatox

https://projects.au.dk/almass

Tarazona et al. 2024 *Environ. Intern.* 191: 108999